Tech pack: Pack 5200 physical test kit

BL-038: order of work, shopping list, guides

Item	Value
generated by	designs/pack_5200/physical_test_kit.py

Pack 5200 physical test kit (BL-038)

Manufacturing-engineer, 2026-11-13. Generated by designs/pack_5200/physical_test_kit.py from designs/pack_5200/tests_physical/ (TEST_PLAN.md, test_register.csv, CA1_fatigue_rig.md). Re-run the script after the register changes; do not edit these files by hand.

Start here

1. Read G1 today: order the long-lead parts and print the frame coupons (they need 7 days).

2. Buy the rest from shopping_list.md (total USD 2,343).

3. Run the sessions G2 to G7 in order. Each guide is one session, written for a phone.

4. Fill results_template.csv (columns measured, date, photo, notes) and send it after each session.

What gets released, in this order

Frame release R1: H1 print files released for load-carrying prints (DECISIONS BL-027: MT-1 cinch, CA-1, T1/T2 gauges; CA1 doc: with IM3-EC and CA-5).

Tests: MT-1, CA-1, IM3-EC, CA-5, IM3-EP, W-PULL, T9, T1, T2, MT-1c

Bag release, step 1: cut and sew: every sewn load point and hardware item proven before the bag fabric is cut.

Tests: PT-1, MT-2, MT-3, US-BS, PT-2, PT-3, PT-5, PT-6, LL-H, B-28 slip, PORT-1, TL-1, FS-O5, FRAY-1, WP-1

Bag release, step 2: batch: first-article bag checks before more bags are cut (release_tests.csv).

Tests: WP-3, WP-2, T9-R3a, T9-R3c, T9-R3b, T9-R3d, GT-1, PT-7, WP-4, WP-5

BL-037 weight gates: the -119.1 g A_bl037 picks (weight_push.md).

Tests: SL-1, CS-1, BP-1, AB-1, BU-1, HS-1, ST-1, HB-1, HB-D2

Field release and the rest: first loaded carry, meat haul, belt and comfort tests.

Tests: HB-7, HB-6, CA-3, CA-4, HB-5, T6, T7, CA-2, HB-4, HB-T, PS-1, BP-2, HB-2, BP-3, RF-1, T3, CR-1, SP-1, T4a, T4b, T4c, T5, T5f, T2u, HB-3, HB-D1, HB-D3, FT-S2, MH-1, MH-2, MH-3, R28-1

Frame release R1 needs, in order: CA-1 at the release level 0.276 (governing flip 0.2556 + 0.02, from the BL-027 body and the 28 mm IM3 cheek; confirmed by BL-039), IM3-EC and CA-5 on the coupons; MT-1 on the bench; then on the first frames IM3-EP and W-PULL (P1), T9, T1, T2 and MT-1c (P2). MT-1c's measured cinch swing sets the CA-1 level R1 needs (BL-039): swing <= 0.4 %: 0.276; up to 70 %: the 0.40 pass as well; above: 0.404 (H1's fatigue with the cinch fully cycling is 0.782 against 1.175 static).

Sessions

session	what	when	hands-on h	machine h
G1	Order the long-lead parts, print and condition the coupons	week 1, day 1	0.0	0
G2	Pull bench weekend: hooks, cams, bartacks, ladder locks	weekend 1 (week 1)	20.0	5
G3	Wash and look: seams without binding	weekend 1, evenings	3.0	9
G4	Lever rig: frame coupons (weeks 2-5)	build weekend 2, then runs day and night	34.5	487

session	what	when	hands-on h	machine h
G5	Bench session 2: fabric, belt and back coupons	any weekend while the lever rig runs (weeks 2-4)	9.9	8
G6	First-article frames on the stand (P1 then P2)	after G4's CA-1, IM3-EC and CA-5 pass (about week 5-6)	52.2	37
G7	Complete pack: fit, field and first-article bag checks	after T1-T4 pass	24.9	2
	total		144.5	548

Hands-on and machine hours are the register's estimates (ASSUMPTION per row). The calendar is set by the lever rig (about 8 CA-1 runs, 3 IM3-EC, CA-5 and CA-3/CA-4 at 1 Hz, ~28 h each) and by the 4-week lab turnaround for AB-1.

Order of work

Full list: order_of_work.csv. Inside each session the R1 tests run first, then the bag, then the BL-037 gates (tests that share a rig or a specimen batch stay together).

#	session	test	releases
1	G1	CA-1 (print only)	Frame release R1
2	G1	IM3-EC (print only)	Frame release R1
3	G1	CA-5 (print only)	Frame release R1
4	G1	CA-3 (print only)	Field release and the rest
5	G1	CA-4 (print only)	Field release and the rest
6	G1	HB-6 (print only)	Field release and the rest
7	G1	T7 (print only)	Field release and the rest
8	G2	MT-1	Frame release R1 + Bag release, step 1: cut and sew
9	G2	PT-1	Bag release, step 1: cut and sew
10	G2	MT-2	Bag release, step 1: cut and sew
11	G2	MT-3	Bag release, step 1: cut and sew
12	G2	US-BS	Bag release, step 1: cut and sew
13	G2	PT-2	Bag release, step 1: cut and sew
14	G2	PT-3	Bag release, step 1: cut and sew
15	G2	PT-5	Bag release, step 1: cut and sew
16	G2	PT-6	Bag release, step 1: cut and sew
17	G2	LL-H	Bag release, step 1: cut and sew
18	G2	B-28 slip	Bag release, step 1: cut and sew
19	G2	PORT-1	Bag release, step 1: cut and sew
20	G2	TL-1	Bag release, step 1: cut and sew
21	G2	FS-O5	Bag release, step 1: cut and sew
22	G2	WP-3	Bag release, step 2: batch
23	G2	SL-1	BL-037 weight gates
24	G2	HB-7	Field release and the rest
25	G3	FRAY-1	Bag release, step 1: cut and sew

#	session	test	releases
26	G3	WP-1	Bag release, step 1: cut and sew
27	G4	CA-1	Frame release R1
28	G4	IM3-EC	Frame release R1
29	G4	CA-5	Frame release R1
30	G4	CS-1	BL-037 weight gates
31	G4	BP-1	BL-037 weight gates + Field release and the rest
32	G4	HB-6	Field release and the rest
33	G4	CA-3	Field release and the rest
34	G4	CA-4	Field release and the rest
35	G4	HB-5	Field release and the rest
36	G4	T6	Field release and the rest
37	G4	T7	Field release and the rest
38	G4	CA-2	Field release and the rest
39	G5	WP-2	Bag release, step 2: batch
40	G5	AB-1	BL-037 weight gates
41	G5	HB-4	Field release and the rest
42	G5	HB-T	Field release and the rest
43	G5	PS-1	Field release and the rest
44	G5	BP-2	Field release and the rest
45	G5	HB-2	Field release and the rest
46	G6	IM3-EP	Frame release R1
47	G6	W-PULL	Frame release R1
48	G6	T9	Frame release R1 + Bag release, step 2: batch
49	G6	T9-R3a	Bag release, step 2: batch
50	G6	T9-R3c	Bag release, step 2: batch
51	G6	BP-3	Field release and the rest
52	G6	RF-1	Field release and the rest
53	G6	T1	Frame release R1
54	G6	T3	Field release and the rest
55	G6	T2	Frame release R1
56	G6	MT-1c	Frame release R1
57	G6	CR-1	Field release and the rest
58	G6	BU-1	BL-037 weight gates
59	G6	SP-1	Field release and the rest
60	G6	T4a	Field release and the rest
61	G6	T4b	Field release and the rest
62	G6	T4c	Field release and the rest

#	session	test	releases
63	G6	T5	Field release and the rest
64	G6	T5f	Field release and the rest
65	G6	T2u	Field release and the rest
66	G7	T9-R3b	Bag release, step 2: batch
67	G7	T9-R3d	Bag release, step 2: batch
68	G7	GT-1	Bag release, step 2: batch
69	G7	PT-7	Bag release, step 2: batch
70	G7	WP-4	Bag release, step 2: batch
71	G7	WP-5	Bag release, step 2: batch
72	G7	HS-1	BL-037 weight gates
73	G7	ST-1	BL-037 weight gates
74	G7	HB-1	BL-037 weight gates + Field release and the rest
75	G7	HB-D2	BL-037 weight gates + Field release and the rest
76	G7	HB-3	Field release and the rest
77	G7	HB-D1	Field release and the rest
78	G7	HB-D3	Field release and the rest
79	G7	FT-S2	Field release and the rest
80	G7	MH-1	Field release and the rest
81	G7	MH-2	Field release and the rest
82	G7	MH-3	Field release and the rest
83	G7	R28-1	Field release and the rest

Value of the gram tests (re-scored, TEST_PLAN §5)

rank	test	grams	g per $-eq	note
1	BP-1	49.9	2.67	BL-005 strip with BP-2 (44.6) + BL-037 K-1 2 mm core
2	FT-S2	101.6	2.26	owner's wing removal (BL-023)
3	CS-1	54.2	1.89	BL-037 B-1 + H-1 (with HB-1 / HS-1)
4	HB-D3	12.6	1.68	BL-006 f1 + f2
5	HB-1	77.2	1.29	front-strap pad (42.9) + BL-037 B-1, B-3 (with CS-1)
6	WP-3	5.4	1.20	protects A-2 (already in B.2)
7	WP-1	36.4	1.04	protects B-1 (already in B.2)
8	WP-5	12.8	1.02	protects B-10 (already in B.2)
9	HB-D1	22.4	1.00	BL-006 b
10	ST-1	6.0	0.98	BL-037 H-2
11	HB-D2	19.5	0.56	BL-006 c2 + BL-037 B-2 0.5 mm
12	HS-1	22.3	0.46	BL-037 H-1

rank	test	grams	g per $-eq	note
13	SL-1	15.0	0.46	BL-037 G-1 + X-1
14	WP-2	6.2	0.40	C-6 lash tabs
15	HB-5	25.0	0.24	package C plate; also a safety gate
16	BU-1	4.0	0.16	BL-037 F-3
17	RF-1	7.2	0.13	C-8 rails
18	CA-1	9.9	0.09	option F-1 only: CA-1 at 0.40 AND MT-1c static AND a buckling re-run (BL-037); not booked. CA-1 stays the top safety gate
19	FS-O5	7.0	0.08	C-5 floor (alternate set); needs PT-3 on ULTRA too
20	AB-1	22.5	0.07	BL-037 G-2 (Martindale lab fee dominates)

Results sheet

results_template.csv has 223 rows over 76 tests. Each row says how it is judged (check: max, min, band, yes, record) against limit. Type numbers in the unit shown; type Y or N for yes/no rows. Do not change the other columns. Read back with:

python -m huntkit.manufacturing.test_results outputs/release/test_kit/results_template.csv --md report.md

It reports each test as PASS / FAIL / PARTIAL / OPEN, each release, and each ASSUMPTION as VERIFIED / FAILED / OPEN (settles column).

Files

- guides/G1.md ... guides/G7.md: one session each
- shopping_list.md, shopping_list.csv
- order_of_work.csv
- results_template.csv
- sketches/.png: rig sketches
- test_kit.pdf: everything above in one PDF
- manifest.json

Shopping list (BL-038)

Total USD 2,342.64. Prices from material records, cited shop pages (links), or marked ASSUMPTION (check at purchase). Cited or record-priced: USD 1,028.64; ASSUMPTION: USD 1,314.00. Prices checked 2026-11-10 (TEST_PLAN §4) and 2026-11-13.

Order these first (long lead):

- cam buckle 25 mm aluminium (Austere type) (MT-3, US-BS): LONG LEAD: RBTR all variants sold out, Extremtextil stock problems (2026-11-13). Order now; buy the pack's 5 cams in the same order

- YKK #10 coil slider, metal (cord pull fitted) (SL-1): LONG LEAD: Rockywoods listed sold out (2026-11-13); Shelby Outdoor EU stocks the YKK 10C slider

- ROHACELL 71 IG-F core for the 2 mm strip (BP-1): LONG LEAD: 2 mm is not a stock sheet (3 mm thinnest, back_system_integration.md); sand a 3 mm offcut or special order

- HyperD 300 printable, custom camo print (AB-1, FRAY-1): LONG LEAD: custom print, lead time not published

- Martindale abrasion test, accredited lab (2 samples + order fee) (AB-1): LONG LEAD: about 4 weeks after the samples arrive

- sacrificial P1 frame (IM3-EP, W-PULL, then T2u) (IM3-EP, W-PULL, T2u): print time: allow 2 weeks with P2

kit	USD
Kit A pull bench	224.08
Kit B lever rig	317.50
Kit C frame stand	227.39
Kit D measuring	140.46
Specimens and lab	1,433.21
total	2,342.64

Not in this list: the bag, harness and belt materials themselves (pack BOM, outputs/release/pack5200_corrie_f_rev3/manufacturing/bom.csv). Swatches are cut from that fabric order, so buy the bag fabric with the kit. You already own (kit E): washing machine, winter gloves, freezer, hose, tape, paint pen, phone.

Kit A pull bench

item	qty	USD	source	for
crane scale 300 kg / 660 lb, 0.1 kg, tare	1 each	59.00	Fruugo US	all pulls
come-along, 2 ton cable puller (Big Red TR8021)	1 each	30.30	Home Depot	all pulls
ratchet straps 1-1/2 in x 10 ft, 4-pack	1 pack	17.99	Harbor Freight 61303	rigging (never above 151 kg)
hardened dowel pins M5 x 22, 50-pack	1 pack	11.79	Zoro	pin fixture
anchors: 4x4 lag-screwed into studs (or tow hitch / tree)	1 set	40.00	lumber yard; ASSUMPTION (shop price, check at purchase)	bench
pin fixture: steel flat bar 40 x 6 x 300, M8 bolts	1 set	20.00	hardware store; ASSUMPTION (shop price, check at purchase)	bench
bar clamps: 2 pairs Al 150 x 25 x 10, M6 bolts, grit cloth	1 set	15.00	hardware store; ASSUMPTION (shop price, check at purchase)	PT-2, PT-3, FS-O5
safety: glasses, gloves, moving blanket, back-up sling	1 set	20.00	hardware store; ASSUMPTION (shop price, check at purchase)	all
calibration jug 20 L	1 each	10.00	hardware store; ASSUMPTION (shop price, check at purchase)	scale check

Kit B lever rig

item	qty	USD	source	for
S-type load cell 100 kg	1 each	23.60	Makerfabs	CA-*, IM3-EC, MT-1c
HX711 load-cell amplifier (SparkFun)	1 each	11.50	SparkFun	CA-*, MT-1c
Arduino Nano Every	1 each	15.81	Arduino Store (EUR 14.50)	CA-*, MT-1c, FSR strips
gear motor 12 V 60 W 60 rpm 8 N m worm (uxcell)	1 each	54.99	eBay	CA-*
12 V 10 A power supply	1 each	20.00	electronics shop; ASSUMPTION (shop price, check at purchase)	CA-*
MOSFET switch, hall sensor + magnets, AS5600 encoder	1 set	15.00	electronics shop; ASSUMPTION (shop price, check at purchase)	CA-*
6061-T6 flat bar 40 x 10 x 600 (lever)	1 each	18.00	metal supplier; ASSUMPTION (shop price, check at purchase)	CA-*
2 x 608 bearings, M8 shoulder bolt, 2 rod ends, M8 rod	1 set	25.00	hardware store; ASSUMPTION (shop price, check at purchase)	CA-*
compression spring 10-12 N/mm, >= 350 N, washer cage	1 each	10.00	hardware store; ASSUMPTION (shop price, check at purchase)	CA-*
turnbuckle M8, Al grip plates 4 x 50 x 40 x 6, M6 bolts	1 set	20.00	hardware store; ASSUMPTION (shop price, check at purchase)	CA-*
plywood 18 mm + 2x4, screws (rig base)	1 set	20.00	lumber yard; ASSUMPTION (shop price, check at purchase)	CA-*
S-type load cell 200 kg (CA-5 static pulls)	1 each	23.60	Makerfabs (200 kg option)	CA-5
40 C box: cooler + 60 W heater + thermostat	1 set	35.00	hardware store; ASSUMPTION (shop price, check at purchase)	CA-3, CA-4, CS-1
3-point bend fixture	1 set	10.00	hardware store; ASSUMPTION (shop price, check at purchase)	CA-3, BP-1
SD-card logger module + USB power bank (MT-1c, logs while the rig moves)	1 set	15.00	electronics shop; ASSUMPTION (shop price, check at purchase)	MT-1c

Kit C frame stand

item	qty	USD	source	for
hip log (4x4 in closed-cell foam to ~36 in)	1 each	10.00	hardware store; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1)	T1-T5
shoulder bar	1 each	10.00	hardware store; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1)	T1-T5
snap shackle, 316 swivel (drop release)	1 each	14.39	Better Boat	T4a, MT-1c
sand tube bags 60 lb	8 each	48.00	hardware store; ASSUMPTION (shop price, check at purchase)	T1-T5, T9, MT-1c, T2u
rubber gym mat 10-20 mm	1 each	30.00	sports shop; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1)	T4
19 mm plywood sheet	1 each	40.00	lumber yard; ASSUMPTION (shop price, check at purchase)	T4c
T5 crank: 12 V wiper motor, supply, crank, bungee, counter	1 set	70.00	auto parts; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1)	T5, MT-1c

item	qty	USD	source	for
22 mm hardwood dowel	1 each	5.00	hardware store; ASSUMPTION (shop price, check at purchase)	SP-1

Kit D measuring

item	qty	USD	source	for
dial indicator 0-1 in with magnetic base (iGaging 34-101)	1 set	39.95	Penn Tool	CA-5, T1, T4b
digital caliper 6 in	1 each	9.99	Harbor Freight 63586	CA-*, CS-1
force-sensitive resistors (Adafruit 166)	8 each	31.60	Adafruit	HB-1, HS-1, RF-1, BP-3
10x loupe	1 each	8.00	hardware store; ASSUMPTION (shop price, check at purchase)	FRAY-1, WP-1, CA-*
conditioning box + magnesium nitrate	1 set	15.00	chemical supplier; ASSUMPTION (shop price, check at purchase)	CA coupons
pocket scale 200 g x 0.01 g	1 each	13.30	Ward's Science (United Scientific)	SL-1, T9-R3d
spring scale 15 kg (147 N), Kern R100150-S-W	1 each	7.62	aio.lv (EUR 6.99)	T9-R3b, GT-1
compression-set jig: 2 Al plates 100 x 100 x 6, 3.0 + 2.5 mm spacers, 4 M6	1 set	10.00	hardware store; ASSUMPTION (shop price, check at purchase)	CS-1
stiff nylon scrub brush + 2 kg weight	1 set	5.00	hardware store; ASSUMPTION (shop price, check at purchase)	AB-1
phyphox app	1 each	0.00	free	HB-2, T4a, GT-1

Specimens and lab

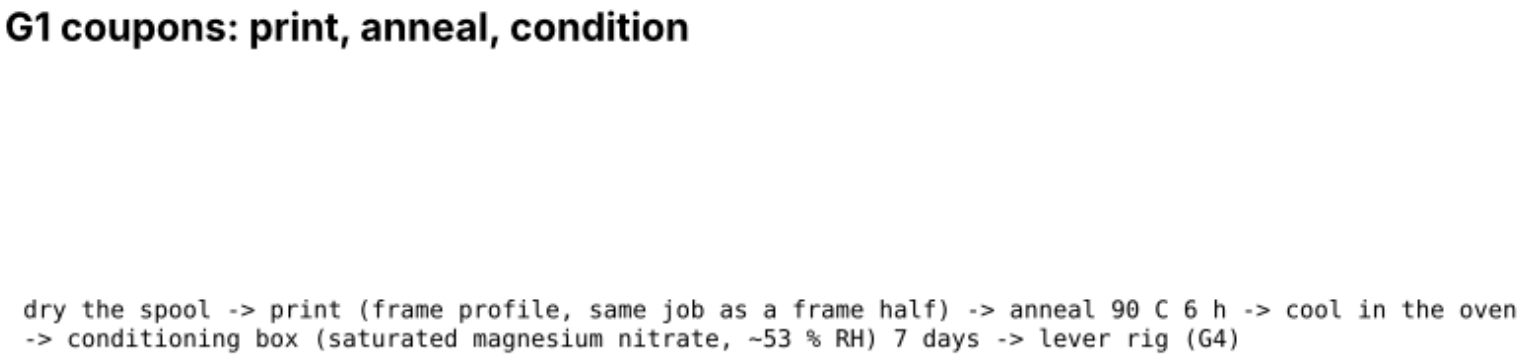
item	qty	USD	source	for				
wiregate carabiner, CAMP Nano 22 class	15 each	150.00	CAMP dealers (rack of 6); record carabiner_wiregate_22kn (cost is an ASSUMPTION there)	PT-1 (5), MT-2 (5), PT-6 (5)				
G-hook 25 mm aluminium, AustriAlpin DY25K-G	5 each	20.00	AustriAlpin / Extremtextil; record g_hook_25mm_aluminium (cost ASSUMPTION there)	MT-2				
cam buckle 25 mm aluminium (Austere type) LONG LEAD	5 each	127.50	Ripstop by the Roll / Extremtextil; record cam_25mm_al	MT-3, US-BS				
tri-glide 25 mm stainless	3 each	8.52	Extremtextil art. 70460; record triglide_25mm (stainless line, EUR 3.10 incl. VAT)	US-BS				
heavy ladder lock 25 mm (ITW Super Ladderloc)	11 each	23.10	Liberty Mountain 6-pack / Buckleguy; record ladder_lock_25mm_heavy	LL-H (5), HB-7 (5), PT-5 (1)				
tri-glide 25 mm nylon	6 each	2.22	Extremtextil; record triglide_25mm	LL-H (5), PT-5 (1)				
D-ring 25 mm aluminium	5 each	9.25	Extremtextil; record d_ring_25mm_aluminium	HB-7				

item	qty	USD	source	for				
YKK #10 coil slider, metal (cord pull fitted) LONG LEAD	5 each	9.95	Rockywoods; alt. Shelby Outdoor EU; record slider_10	SL-1				
#10 coil zipper chain	0.4 m	1.60	Ripstop by the Roll; record coil_zipper_10	SL-1				
Dyneema cord 3 mm (slider pulls)	1 m	0.90	Ripstop by the Roll; record dyneema_cord_3mm	SL-1				
printed label (one, heat transfer)	1 each	5.00	label printer; ASSUMPTION (shop price, check at purchase)	SL-1				
side-release buckle 25 mm	2 each	1.60	Strapworks; record buckle_side_release_25mm	ST-1				
poly webbing 25 mm	20 m	14.00	Strapworks; record poly_webbing_25mm	MT-1, MT-3, US-BS, LL-H, HB-7, PT-5, PT-6, PORT-1, GT-1				
poly webbing 38 mm	3 m	3.00	Strapworks; record poly_webbing_38mm	PT-6 links, WP-3				
XLPE L200 foam 5 + 6 mm	0.35 m2	5.25	Foam By Mail; record xlpe_foam_l200	CS-1 (0.25), HS-1 (0.1)				
EVA 45C foam 5 mm	0.1 m2	2.50	Seattle Fabrics; record eva_foam_45c	HS-1				
HDPE sheet 0.5 mm (BL-037 stiffener wing)	0.05 kg	0.30	McMaster-Carr; record hdpe_sheet	HB-D2				
ROHACELL 71 IG-F core for the 2 mm strip LONG LEAD	0.02 m2	2.40	Rock West Composites; record rohacell_71_igf	BP-1				
200 g 2x2 carbon twill + epoxy for the 2 mm BP-1 coupon skins	0.028 m2	1.26	left over from the spreader / strip layup (else any composites shop); record cfrp_t300_3k_woven_epoxy_ply (BL-039)	BP-1				
HyperD 300 printable, custom camo print LONG LEAD	1 yd	30.25	Ripstop by the Roll (custom print, 1 yd min); record polyester_300d_printable	AB-1, FRAY-1				
Martindale abrasion test, accredited lab (2 samples + order fee) LONG LEAD	1 order	249.61	Measurlabs (EUR 66 / sample + EUR 97 fee)	AB-1				
specimens for PT-2: 5 swatches built like the pack, B-17 cut 70 mm tall	1 lot	15.00	see register; test_register.csv PT-2: ASSUMPTION Hypalon + laminate offcuts	PT-2				
specimens for PT-3: 5 swatches without binding (B.2 as sewn)	1 lot	15.00	see register; test_register.csv PT-3: ASSUMPTION offcuts	PT-3				
specimens for WP-1: 4 seam pairings 330D	330D, 330D	ULTRA 200, 500D	330D, 100D	ULTRA; 301 + 2nd row	1 lot	5.00	see register; test_register.csv WP-1: offcuts	WP-1

item	qty	USD	source	for				
specimens for CA-1: 12 XY + 4 Z + 3 spare bars, printed and annealed like the frame, 7 days at 23 C	1 lot	15.00	see register; test_register.csv CA-1: CA1_fatigue_rig.md §5 (filament $80/kg)	CA-1				
specimens for IM3-EC: 3 eye coupons: 120 mm stay segment round z 520 with the IM3 eye and the 28 mm ch	1 lot	10.00	see register; test_register.csv IM3-EC: ASSUMPTION: filament + 3 tongues	IM3-EC				
specimens for WP-2: 5 laser-cut B-11 tabs	1 lot	5.00	see register; test_register.csv WP-2: ASSUMPTION	WP-2				
specimens for FS-O5: 5 warp + 5 weft strips 50 mm wide (ISO 13934-1 / ASTM D5035 style) + 5 PT-3 swat	1 lot	35.00	see register; test_register.csv FS-O5: ASSUMPTION fabric $25 + PT-3 set $10	FS-O5				
specimens for HB-6: 5 inserts in printed, annealed PAHT-CF blocks	1 lot	5.00	see register; test_register.csv HB-6: ASSUMPTION	HB-6				
specimens for CA-5: 4 stub pairs, one stay, 60 mm each side of z 234, real bosses, Ti bolts	1 lot	40.00	see register; test_register.csv CA-5: frame_comp_verdict.md §6 (ASSUMPTION)	CA-5				
specimens for CA-3: 3 Z bars + 3 short-beam blocks 24 x 8 x 6	1 lot	5.00	see register; test_register.csv CA-3: ASSUMPTION	CA-3				
specimens for CA-4: 2 XY bars	1 lot	10.00	see register; test_register.csv CA-4: frame_comp_verdict.md §6 (ASSUMPTION)	CA-4				
specimens for CA-2: 3 XY + 3 Z bars per anneal	1 lot	10.00	see register; test_register.csv CA-2: ASSUMPTION	CA-2				
specimens for HB-4: 1 ear stitched into a wing-root coupon (3 recommended)	1 lot	15.00	see register; test_register.csv HB-4: ASSUMPTION plate offcut + foam	HB-4				
specimens for HB-T: 5 tabs sewn like the belt: 50 mm webbing round a buckle bar, 25 mm anchor and lo	1 lot	5.00	see register; test_register.csv HB-T: ASSUMPTION webbing	HB-T				
specimens for TL-1: 5 loops: 25 mm webbing x 170 mm + POM toggle, on an 18 mm CF tube offcut (no wra	1 lot	8.00	see register; test_register.csv TL-1: ASSUMPTION toggles + tube offcut	TL-1				
specimens for HB-5: 1 plate on the Ti pin over a foam block	1 lot	45.00	see register; test_register.csv HB-5: ASSUMPTION plate $25 + fixture $20	HB-5				
specimens for PS-1: pad seat on a printed pan offcut	1 lot	5.00	see register; test_register.csv PS-1: ASSUMPTION	PS-1				

item	qty	USD	source	for				
specimens for BP-2: 100 mm printed H1 stay segment (z about 260) on a strip offcut, 10 mm foam under	1 lot	5.00	see register; test_register.csv BP-2: ASSUMPTION carbon paper; Prescale roll about EUR 1275 not bought (§4.4)	BP-2				
specimens for T6: 3 corner coupons, 0.8 mm glass-bead bondline	1 lot	30.00	see register; test_register.csv T6: ASSUMPTION tube offcuts + MMA	T6				
specimens for T7: 150 mm printed H1 sleeve segment + UHMWPE bushing + leg offcut	1 lot	10.00	see register; test_register.csv T7: ASSUMPTION	T7				
specimens for SP-1: P2 spreader	1 lot	5.00	see register; test_register.csv SP-1: ASSUMPTION dowel	SP-1				
specimens for HB-D2: belt B wings side by side: 0.5 mm and 0.75 mm HDPE stiffener (and the 1.0 mm ref	1 lot	5.00	see register; test_register.csv HB-D2: ASSUMPTION 1.0 mm HDPE offcut	HB-D2				
specimens for RF-1: rails on P2, 3 testers	1 lot	10.00	see register; test_register.csv RF-1: ASSUMPTION foam	RF-1				
specimens for WP-5: ports, hose frames, cam-cup base	1 lot	5.00	see register; test_register.csv WP-5: ASSUMPTION	WP-5				
sacrificial P1 frame (IM3-EP, W-PULL, then T2u) LONG LEAD	1 each	437.00	owner's print + parts; frame_h1.md 5 unit cost	IM3-EP, W-PULL, T2u				
specimens for BU-1: 2 UHMWPE crown bushings cut 50 mm from the 70 mm stock, on P2	1 lot	10.00	see register; test_register.csv BU-1: ASSUMPTION spare UHMWPE tube length $10 (uhmwpe_sheet record has no tube price)	BU-1				

G1: Order the long-lead parts, print and condition the coupons



When: week 1, day 1

Rig: 3D printer (frame profile), oven, conditioning box (kit D)

Kits: B (filament), D

Start here, today. The frame coupons need 7 days in the conditioning box before the lever rig can test them, so they set the pace for frame release R1. Order the long-lead parts the same day.

Safety (read before you start)

- Hot oven and hot nozzle: gloves. Magnesium nitrate is an oxidizer and an irritant: gloves, keep it away from heat sources and children, label the box.

Rig sketch

dry the spool -> print (frame profile, same job as a frame half) -> anneal 90 C 6 h -> cool in the oven

-> conditioning box (saturated magnesium nitrate, ~53 % RH) 7 days -> lever rig (G4)

Picture: sketches/G1_rig.png

Tests in this session, in order

#	test	releases
1	CA-1	Frame release R1
2	IM3-EC	Frame release R1
3	CA-5	Frame release R1
4	CA-3	Field release and the rest
5	CA-4	Field release and the rest
6	HB-6	Field release and the rest
7	T7	Field release and the rest

Print list (one job, same spool and profile as the frame)

coupon	for	count
XY dog-bone bars 170 x 20 (gauge 10 x 4 x 60)	CA-1	12 + 2 spare
Z bars 100 x 20 (gauge 10 x 6 x 30), standing	CA-1	4 + 1 spare
Z bars + short-beam blocks 24 x 8 x 6	CA-3	3 + 3
XY bars	CA-4	2
z 234 joint stub pairs (H1 bosses, traps)	CA-5	4
IM3 eye coupons, 120 mm stay at z 520, 28 mm cheek	IM3-EC	3
insert blocks (M5 heat-set)	HB-6	5
150 mm H1 sleeve segment	T7	1

Steps:

1. Dry the spool. Print with the frame profile (designs/pack_5200/print/frame_h1/README.md): 0.6 mm hardened nozzle, 290 C, bed 100 C, 0.20 mm layers, 6 walls, 100 % infill.

2. XY bars lie flat on their 10 mm face (4 mm is the print height), like the frame.

3. Anneal 90 C for 6 h on a flat plate; cool in the oven.

4. Into the conditioning box for 7 days (23 C, about 53 % RH). Write the date on the box.

5. Measure each bar's gauge width and thickness at 3 points with the caliper; write the smallest on the bar.

Order today (long lead)

- 25 mm aluminium cams (MT-3, US-BS) and the pack's own 5: both resellers out of stock on 2026-11-13.

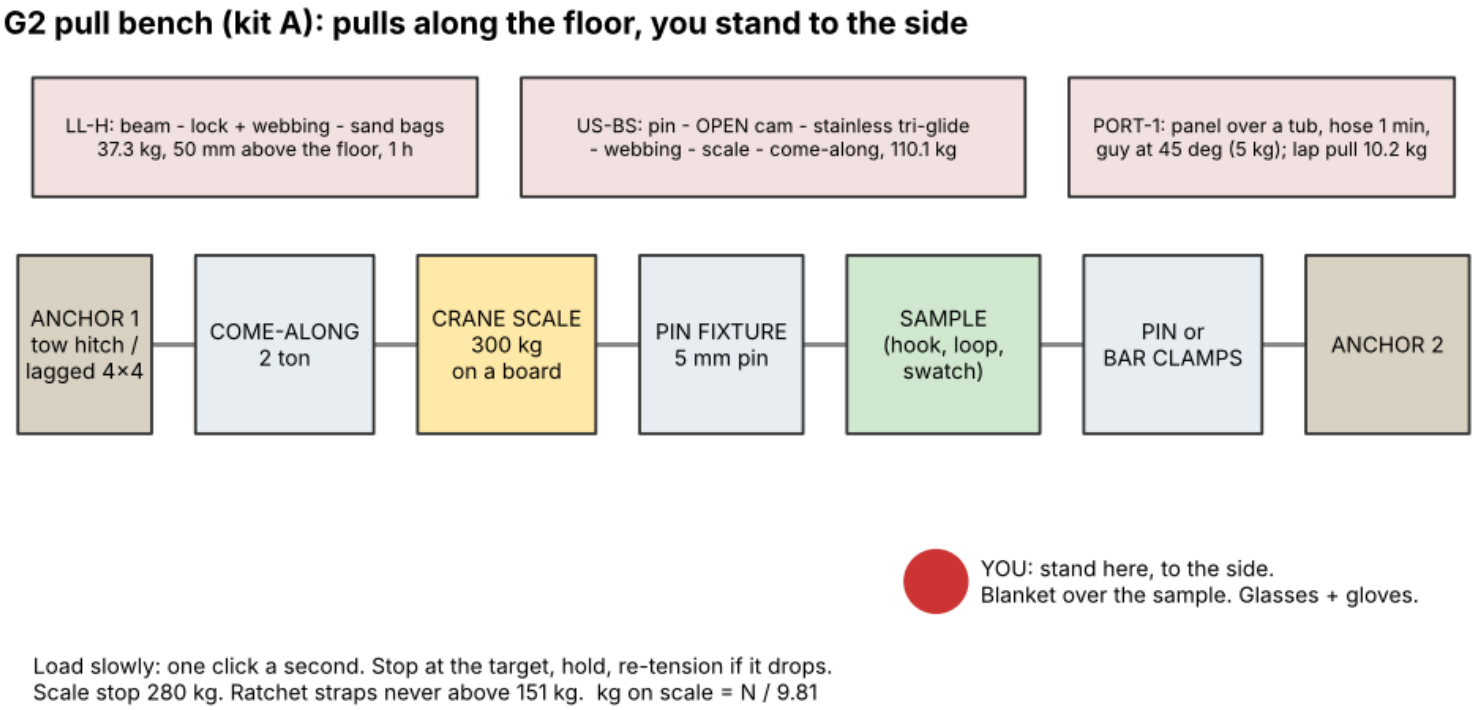
- YKK #10 sliders (SL-1): Rockywoods sold out; Shelby Outdoor EU has them.

- HyperD 300 custom camo print (AB-1, FRAY-1) and the Martindale lab booking (about 4 weeks).

- 2 mm ROHACELL 71 (BP-1): not a stock sheet; sand a 3 mm offcut or special-order.

- Everything else in shopping_list.md.

G2: Pull bench weekend: hooks, cams, bartacks, ladder locks



When: weekend 1 (week 1)

Rig: kit A pull bench (come-along, 300 kg crane scale, 5 mm pin fixture, bar clamps)

Kits: A, D (caliper, loupe, pocket scale), E

One bench, one weekend. MT-1 goes first because frame release R1 needs it; the rest are every sewn load point on the bag. After this weekend you may cut the bag (if G3's wash also passes).

About 20.0 h hands-on and 5 h of machine time.

Safety (read before you start)

- Stand to the side of the load line, never in line with it. Moving blanket over every sample above 50 kg.

- Safety glasses and gloves on. Nobody near the line while it is loaded.

- Ratchet straps never in the load line above 151 kg (1480 N, ASSUMPTION: a third of the 1000 lb label).
- Crane scale stop: 280 kg (2750 N). PT-1 and PT-2 at 260.2 kg are the heaviest pulls: come-along only.
- LL-H hangs a dead load: nobody under it, 50 mm off the floor so a slip ends on the floor.
- If anything fails: stop, photograph, write it down, send the sheet. Do not repair and continue.

Rig sketch

ANCHOR 1 -- sling -- COME-ALONG -- CRANE SCALE -- PIN FIXTURE -- SAMPLE -- PIN / BAR CLAMP -- sling -- ANCHOR 2

(tow hitch, tree, lagged 4x4) (2 ton) (300 kg, on a board) (5 mm hardened dowel pin) (5 mm pin, or the bar clamps for swatches)

You stand HERE, to the side --> o (never in line with the pull; blanket over the sample)

LL-H dead hang: beam -- lock + webbing -- sand bags 37.3 kg, 50 mm above the floor

US-BS: pin -- [open cam] -- [stainless tri-glide] -- webbing -- scale -- come-along

Picture: sketches/G2_rig.png

Sew before the weekend (about 5 h, same machine, thread and needle as the pack):

1. MT-1: 10 loops of 25 mm poly webbing; fold each end into a 20 mm loop and close it with a 25 x 25 mm box-X, stitch 301, 7 SPI, bonded nylon tex 135, Nm 140 round-point needle.
2. PT-2: 5 x T3 tongue on a 70 mm B-17 tab on a B-01 swatch about 200 x 200 mm (bag_attachment.md 5 T3).
3. PT-3: 5 x HE eye on a B-18 + B-01 + B-05 seam swatch, no binding.
4. PT-6: 5 B-24 links to the cut list (carabiner + 3 loops).
5. US-BS: 3 x 600 mm of 25 mm webbing threaded through a cam with the stainless tri-glide behind it.
6. LL-H: 5 x 25 mm webbing through a heavy ladder lock + nylon tri-glide; 2 of them on a hinge-band tab like the B-28 front lock.
7. PORT-1: one 300 x 300 mm B-01 panel with a B-27 lapped slit and B-27f facing (op sequence).
8. SL-1: a 400 mm #10 coil door sample on 330D tabs with 2 sliders; a 3 mm cord pull on each of 5 sliders.
9. Label every sample with a paint pen: test ID and sample number (PT-2-1 ... PT-2-5).

Tests in this session, in order

#	test	releases
1	MT-1	Frame release R1 + Bag release, step 1: cut and sew
2	PT-1	Bag release, step 1: cut and sew
3	MT-2	Bag release, step 1: cut and sew
4	MT-3	Bag release, step 1: cut and sew
5	US-BS	Bag release, step 1: cut and sew
6	PT-2	Bag release, step 1: cut and sew

#	test	releases
7	PT-3	Bag release, step 1: cut and sew
8	PT-5	Bag release, step 1: cut and sew
9	PT-6	Bag release, step 1: cut and sew
10	LL-H	Bag release, step 1: cut and sew
11	B-28 slip	Bag release, step 1: cut and sew
12	PORT-1	Bag release, step 1: cut and sew
13	TL-1	Bag release, step 1: cut and sew
14	FS-O5	Bag release, step 1: cut and sew
15	WP-3	Bag release, step 2: batch
16	SL-1	BL-037 weight gates
17	HB-7	Field release and the rest

1. MT-1: 25 mm box-X bartack proof at the BL-010 per-row loads

Why: every 25 mm box-X on IM1-IM3 and US1/US2; BL-003 (a) (A_imus)

Specimen: 10 loops: 25 mm poly webbing, box-X 25 x 25, stitch 301, 7 SPI, bonded nylon tex 135

load	N	lb	kg on the scale
IM loops (5)	908	204	92.6
US loops (5)	1080	243	110.1

Steps

1. Sew 10 loops exactly like the pack

2. Loop over the 5 mm pin

3. 5 loops to 908 N (92.6 kg) and 5 to 1080 N (110.1 kg)

4. Hold 1 min each

Pass: 5 of 5 hold 908 N (IM) and 5 of 5 hold 1080 N (US) for 1 min, no broken stitch

Write in the results sheet (results_template.csv, column measured):

- MT-1-01 to MT-1-05 (IM#1 to IM#5): held 92.6 kg (908 N) 1 min, no broken stitch. Type Y or N.

- MT-1-06 to MT-1-10 (US#1 to US#5): held 110.1 kg (1080 N) 1 min, no broken stitch. Type Y or N.

Note: was 910 N on 38 mm loops. If BL-003 (a) fails, the 38 mm IM/US set comes back and the 38 mm box-X at 910 N applies

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8 (BOX25); designs/pack_5200/BUILD_PLAN.md §5a.0

2. PT-1: IM3 bag carabiner on the sewn T3 tongue, per-row proof (BL-010)

Why: IM3 bag hooks; BL-004 REC (A_att, IM2 hooks deleted so IM3 carries the top couple)

Specimen: 5 carabiners from the lot (CAMP Nano 22 class), each on a sewn T3 tongue

load	N	lb	kg on the scale
proof, 60 s	2553	574	260.2

Steps

1. Carabiner over the 5 mm steel pin

2. Sewn T3 tongue on the carabiner

3. Pull with the come-along to 2553 N (260.2 kg on the scale)

4. Hold 60 s

5. Release

6. Check the gate closes and look for set

Pass: 5 of 5 hold 2553 N for 60 s, gate closes after, no visible set

Write in the results sheet (results_template.csv, column measured):

- PT-1-01 to PT-1-05 (#1 to #5): held 260.2 kg (2553 N) 60 s, gate closes, no set. Type Y or N.

Note: was 2120 N on a 38 mm hook (bag_attachment.md); raised by BL-010 (1702 N x 1.5). 260 kg is under the 280 kg scale stop. Fallback if BL-004 REC fails: hood + IM2 hooks (PT-4 still applies)

Defined in: designs/pack_5200/analysis/loadpath_verification.md §5, §8 (CARA_PROOF); designs/pack_5200/analysis/bag_attachment.md §5

3. MT-2: meat-strap hook proof per strap (BL-010)

Why: IM1-IM3 free-end hooks; BL-003 (a) (A_imus)

Specimen: 5 x 25 mm rated G-hooks, 5 x IM3 carabiners (the PT-1 lot may be reused after PT-1 only if no set)

load	N	lb	kg on the scale
G-hooks (5)	645	145	65.7
carabiners (5)	910	205	92.8

Steps

1. As PT-1: G-hooks to 645 N (65.7 kg, covers IM1 410 N), carabiners to 910 N (92.8 kg)

2. Hold 1 min

Pass: 5 of 5 G-hooks hold 645 N and 5 of 5 carabiners hold 910 N for 1 min, gates close

Write in the results sheet (results_template.csv, column measured):

- MT-2-01 to MT-2-05 (G#1 to G#5): G-hook held 65.7 kg (645 N) 1 min, gate closes. Type Y or N.

- MT-2-06 to MT-2-10 (C#1 to C#5): carabiner held 92.8 kg (910 N) 1 min, gate closes. Type Y or N.

Note: BL-038: in the kit the MT-2 carabiners are a separate lot of 5 from PT-1

Defined in: designs/pack_5200/analysis/loadpath_verification.md §5, §8; designs/pack_5200/BUILD_PLAN.md §5a.0

4. MT-3: 25 mm cam buckle slip at the BL-010 per-row loads

Why: IM and US cams; BL-003 (a) (A_imus)

Specimen: 5 x 25 mm Al cams on 25 mm webbing

load	N	lb	kg on the scale
first hold	908	204	92.6
second hold	1080	243	110.1

Steps

1. Thread the cam as in the pack
2. Paint-pen mark at the cam mouth
3. Pull to 908 N (92.6 kg), hold 1 min, measure
4. Then to 1080 N (110.1 kg), hold 1 min, measure

Pass: mark moves < 3 mm at 908 N and at 1080 N, 5 of 5

Write in the results sheet (results_template.csv, column measured):

- MT-3-01 to MT-3-05 (a#1 to a#5): mark moved at 92.6 kg (908 N). Mm, pass if at most 3.
- MT-3-06 to MT-3-10 (b#1 to b#5): mark moved at 110.1 kg (1080 N). Mm, pass if at most 3.

Note: BL-038: price from the record (was ASSUMPTION $4). Cams LONG LEAD (both resellers out of stock 2026-11-13)

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8 (CAM25); designs/pack_5200/BUILD_PLAN.md §5a.0

5. US-BS: US backstop: stainless tri-glide behind an open cam stops the cradle strap

Why: US1/US2 backstop (bag release)

Specimen: 3 x 25 mm poly webbing 600 mm, Al cam (the MT-3 cams, re-used) with the stainless 25 mm tri-glide threaded behind it as the op sequence

load	N	lb	kg on the scale
US proof, cam held open	1080	243	110.1

Steps

1. Thread cam and stainless tri-glide as in the pack
2. Paint-pen mark on the webbing at the tri-glide
3. Hold the cam lever open with a zip tie so it cannot grip
4. Pull slowly with the come-along to 110.1 kg (1080 N)
5. Hold 1 min
6. Measure how far the mark moved

Pass: tri-glide stops the strap within 10 mm and holds 1080 N for 1 min, 3 of 3

Write in the results sheet (results_template.csv, column measured):

- US-BS-01 to US-BS-03 (#1 to #3): strap stopped within. Mm, pass if at most 10.

- US-BS-04 to US-BS-06 (h#1 to h#3): held 110.1 kg (1080 N) 1 min. Type Y or N.

Note: BL-038: added from release_tests.csv. Run after MT-3 on the same cams

Defined in: outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv; data/materials/hardware/triglide_25mm.yaml (stainless alternative); docs/DECISIONS.md 2026-11-13 orchestrator row

6. PT-2: T3 tongue on the 70 mm B-17 tab on a B-01 swatch (BL-010 load)

Why: IM3 bag attachment; BL-004 REC B-17 trim (A_att)

Specimen: 5 swatches built like the pack, B-17 cut 70 mm tall

load	N	lb	kg on the scale
proof, 60 s	2553	574	260.2

Steps

1. Clamp the B-01 swatch 120 mm wide between bar clamps

2. Tongue loop on the pin

3. Pull to 2553 N (260.2 kg)

4. Hold 60 s

Pass: 5 of 5 hold 2553 N, no stitch break, no laminate tear

Write in the results sheet (results_template.csv, column measured):

- PT-2-01 to PT-2-05 (#1 to #5): held 260.2 kg (2553 N) 60 s, no stitch break, no tear. Type Y or N.

Note: was 2120 N on the 110 mm tab

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8; designs/pack_5200/analysis/bag_attachment.md §5; designs/pack_5200/analysis/attachment_diet.md (B-17 110 -> 70 mm)

7. PT-3: HE eye on B-18 + B-01 + B-05 seam, built without binding

Why: hinge eyes; B-1 (no binding); C-5 if on ULTRA 200X

Specimen: 5 swatches without binding (B.2 as sewn)

load	N	lb	kg on the scale
proof, 60 s	1530	344	156.0

Steps

1. 5 mm pin through the eye

2. Clamp the swatch

3. Pull to 1530 N

4. Hold 60 s

Pass: 5 of 5 hold 1530 N

Write in the results sheet (results_template.csv, column measured):

- PT-3-01 to PT-3-05 (#1 to #5): held 156 kg (1530 N) 60 s. Type Y or N.

Note: if the floor goes to ULTRA 200X (C-5), sew a second set of 5 on ULTRA 200X (see FS-O5)

Defined in: designs/pack_5200/analysis/bag_attachment.md §5; designs/pack_5200/analysis/corrie_body.md §12.4 (rebuilt); designs/pack_5200/analysis/fabric_swap_evidence.md O-5

8. PT-5: guy ladder lock + tri-glide keeper slip

Why: TG guys (top anchor)

Specimen: 25 mm webbing, ladder lock + tri-glide keeper

load	N	lb	kg on the scale
10 cycles to	170	38	17.3

Steps

1. Mark webbing

2. 10 cycles 0 -> 170 N -> 0 by hand on the come-along

3. Measure the mark

Pass: slip < 5 mm over 10 cycles

Write in the results sheet (results_template.csv, column measured):

- PT-5-01 (#1): total slip after 10 cycles to 17.3 kg. Mm, pass if at most 5.

Defined in: designs/pack_5200/analysis/bag_attachment.md §5

9. PT-6: B-24 float stand-off link

Why: float mode (D-33)

Specimen: 5 links built to the cut list (hook + 3 loops)

load	N	lb	kg on the scale
proof, 60 s	1410	317	143.7

Steps

1. Link hook on the pin

2. 38 mm loop through the 220 mm loop

3. Pull in line to 1410 N

4. Hold 60 s

Pass: 5 of 5 hold 1410 N, no box-X break, gate closes

Write in the results sheet (results_template.csv, column measured):

- PT-6-01 to PT-6-05 (#1 to #5): held 143.7 kg (1410 N) 60 s, no box-X break, gate closes. Type Y or N.

Defined in: designs/pack_5200/analysis/bag_attachment.md §5

10. LL-H: heavy 25 mm ladder lock slip, 1 h hold (guys and W1-W3)

Why: guy ladder locks on B-22 (BL-004 REC, A_att); W1-W3 float clamp locks; B-28 crown strap locks (the B-28 slip gate, BL-031)

Specimen: 5 heavy ladder locks + tri-glides on 25 mm webbing (2 threaded as a B-28 front lock on a hinge-band tab)

load	N	lb	kg on the scale
dead load, 1 h	366	82	37.3

Steps

1. Thread as in the pack
2. Paint-pen mark at the lock
3. Hang 37.3 kg (366 N, sand bags) 50 mm above the floor for 1 h, all 5 side by side
4. Measure the mark

Pass: slip < 5 mm after 1 h, 5 of 5 (366 N covers the W 338 N proof)

Write in the results sheet (results_template.csv, column measured):

- LL-H-01 to LL-H-05 (#1 to #5): slip after 1 h at 37.3 kg (366 N). Mm, pass if at most 5.

Note: slip criterion as PT-5 (ASSUMPTION: 5 mm); stand clear of the hanging load

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 S9, A3, §5; designs/pack_5200/analysis/attachment_diet.md (guys 366 N)

11. B-28 slip: B-28 crown strap on its heavy ladder lock

Why: B-28 crown straps

Specimen: LL-H specimens (2 of the 5 threaded as a B-28 front lock on a hinge-band tab)

Nothing extra to run: covered by LL-H: same 25 mm heavy lock, 366 N proof above the 288 N envelope (BL-031 dropped the 20 mm gate).

12. PORT-1: B-27 guy port mock-up: hose, 100 N lap pull, gloved lock use

Why: B-27 construction (bag release)

Specimen: 300 x 300 mm B-01 (ULTRA 200) panel with one B-27 lapped slit and its B-27f facing sewn as op sequence, 25 mm guy webbing through it

load	N	lb	kg on the scale
guy at 45 deg during the hose	49	11	5.0
lap pull	100	22	10.2

Steps

1. Clamp the panel over a plastic tub, outside face up, a dry paper towel in the tub

2. Hang 5 kg (49 N) on the guy so it pulls at 45 deg through the port

3. Hose from 1 m above for 1 min

4. Check the towel

5. Then pull the guy straight through the lap to 10.2 kg (100 N) on the crane scale and measure the lap gap

6. With winter gloves thread and adjust the guy ladder lock at the port 5 times

Pass: towel dry; lap gap <= 2 mm at 100 N (ASSUMPTION 'stays closed' = 2 mm); 5 of 5 gloved adjustments

Write in the results sheet (results_template.csv, column measured):

- PORT-1-01 (1): towel dry after 1 min hose. Type Y or N.

- PORT-1-02 (1): lap gap at 10.2 kg (100 N). Mm, pass if at most 2.

- PORT-1-03 (1): gloved lock adjustments OK out of 5. Count, pass if at least 5.

Note: BL-038: added from release_tests.csv / corrie_f_body 13.9

Defined in: designs/pack_5200/analysis/corrie_f_body.md 13.8, 13.9; outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv

13. TL-1: guy toggle loop on an 18 mm tube

Why: BL-004 REC guy loops (A_att)

Specimen: 5 loops: 25 mm webbing x 170 mm + POM toggle, on an 18 mm CF tube offcut (no wrap)

Steps

1. Tube clamped in the pin fixture

2. Loop round the tube, toggle through

3. Pull aft-outward to 366 N (37.3 kg)

4. Hold 60 s

5. Repeat 1 sample after 4 h in the freezer (as PT-4)

Pass: 5 of 5 hold 366 N, toggle does not pull through or crack; cold sample holds; no crush mark on the tube

Write in the results sheet (results_template.csv, column measured):

- TL-1-01 (1): passes the register criterion: 5 of 5 hold 366 N, toggle does not pull through or crack; cold sample holds; no crush mark on the tube. Type Y or N.

Note: the cold repeat is added here (POM is brittler cold), not in the source

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 A3, §8 (GUY_DIR); designs/pack_5200/analysis/attachment_diet.md §7

14. FS-O5: ULTRA 200X 50 mm strip tensile (C-5 / O-5)

Why: C-5 bag bottom 500D -> ULTRA 200X

Specimen: 5 warp + 5 weft strips 50 mm wide (ISO 13934-1 / ASTM D5035 style) + 5 PT-3 swatches on ULTRA 200X

Steps

1. Clamp each strip in the 50 mm bar grips, 200 mm gauge

2. Pull slowly to break

3. Record the peak

Pass: mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm

Write in the results sheet (results_template.csv, column measured):

- FS-O5-01 (1): passes the register criterion: mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm. Type Y or N.

Note: between 1195 and 1840: swap allowed, report to structural

Defined in: designs/pack_5200/analysis/fabric_swap_evidence.md O-5

15. WP-3: gloved tail on W1-W3 ladder locks

Why: A-2 wrap tails 250 -> 150 mm

Specimen: ladder lock on 38 mm webbing with a 150 mm tail

Steps

1. Winter gloves on

2. Release and re-tension, loaded (10 kg hanging) and slack, 5 times each

Pass: release and re-tension in one grab every time

Write in the results sheet (results_template.csv, column measured):

- WP-3-01 (1): passes the register criterion: release and re-tension in one grab every time. Type Y or N.

Note: A-2 is already applied in B.2: a fail adds 5.4 g back

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4

16. SL-1: weigh the cord-pull door sliders and the printed label; gloved door cycles

Why: BL-037 G-1 (-8.0 g) and X-1 (-7.0 g)

Specimen: 5 YKK #10 coil slider bodies with a 3 mm cord pull each; the printed label; a 400 mm #10 coil door sample on 330D tabs with 2 sliders

Steps

1. Fit the cord pull to each slider

2. Weigh each on the 0.01 g scale

3. Weigh the label

4. With winter gloves open and close the door sample 20 times

Pass: each slider with pull <= 15.0 g; label <= 1.0 g; 20 of 20 without a snag

Write in the results sheet (results_template.csv, column measured):

- SL-1-01 to SL-1-05 (#1 to #5): slider + cord pull mass. G, pass if at most 15.0.

- SL-1-06 (label): printed label mass. G, pass if at most 1.0.

- SL-1-07 (door): gloved open / close without a snag, out of 20. Count, pass if at least 20.

Note: BL-038: added (BL-037). Sliders are LONG LEAD (Rockywoods sold out 2026-11-13)

Defined in: designs/pack_5200/analysis/weight_push.md 2 (G-1, X-1), 7 SLIDER-CORD, LABEL

17. HB-7: hipbelt heavy ladder-lock slip, corrected 2:1 routing

Why: belt B 2:1 pull

Specimen: 5 heavy 25 mm ladder locks on 25 mm webbing, routed anchor -> D-ring -> lock

load	N	lb	kg on the scale
hold 1 min, no slip	426	96	43.4
slip must be above	490	110	49.9

Steps

1. Mark webbing

2. Pull the lock leg to 426 N, hold 1 min

3. Then pull on until it slips and record the load

Pass: no slip at 426 N (1.5 x 284 N); slip load >= 490 N keeps the record

Write in the results sheet (results_template.csv, column measured):

- HB-7-01 to HB-7-05 (#1 to #5): no slip at 43.4 kg (426 N), 1 min. Type Y or N.

- HB-7-06 to HB-7-10 (s#1 to s#5): slip load. Kg, pass if at least 50.0.

Note: pass number derived here: 1.5 x 284 N = 426 N (doc gives only 'as the record notes')

Defined in: designs/pack_5200/analysis/hipbelt_structural.md §5, §8

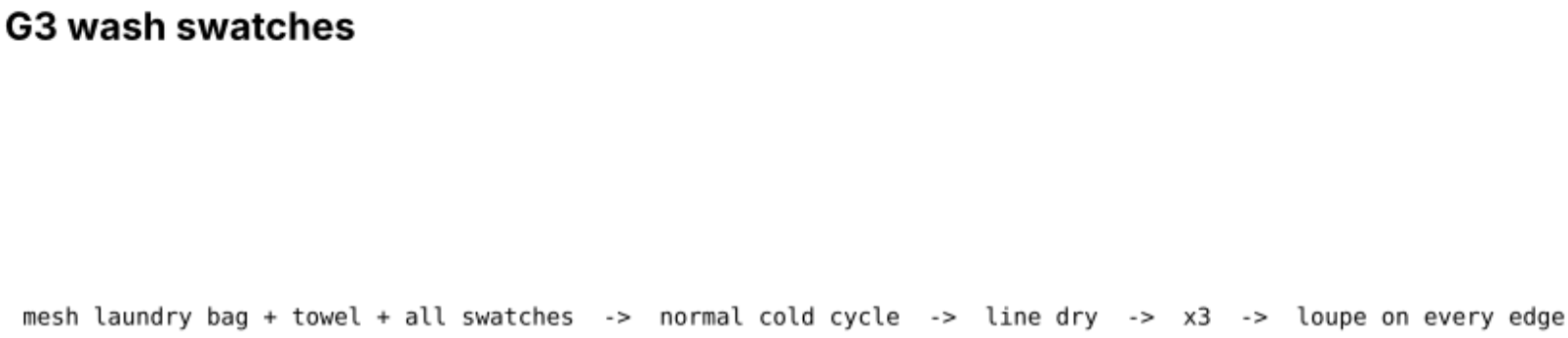
After the session

1. Fill measured and date for every row you ran; leave the rest empty.

2. Photograph every failed or doubtful sample; put the photo number in photo.

3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

G3: Wash and look: seams without binding



When: weekend 1, evenings

Rig: washing machine, mesh laundry bag, 10x loupe

Kits: E, D (loupe)

Both swatch sets go in the same wash. Three cold cycles, line dry, then look at every edge.

About 3.0 h hands-on and 9 h of machine time.

Safety (read before you start)

- Nothing loaded. Keep the hot knife off unless a swatch fails.

Rig sketch

mesh laundry bag + towel + all swatches -> normal cold cycle -> line dry -> x3 -> loupe on every edge

Picture: sketches/G3_rig.png

Sew first: the 6 FRAY-1 SSa2 swatches (rotary-cut, edges NOT seared) and the 4 WP-1 swatches (seared), 200 mm each, as listed under each test below.

Tests in this session, in order

#	test	releases
1	FRAY-1	Bag release, step 1: cut and sew
2	WP-1	Bag release, step 1: cut and sew

1. FRAY-1: wash-swatch fray of the un-bound SSa2 body seam (rev 3 fabrics, as cut)

Why: un-bound body seams (bag release)

Specimen: 6 SSa2 swatches 200 mm long, rotary-cut edges NOT seared (as the prototype cutting plan cuts them): ULTRA 200|330D, 330D|330D, 500D|330D, 630D printable|330D, HyperD 300|330D, 100D|330D; 301, 8 SPI, tex 90, Nm 110 round point

Steps

1. Sew each swatch as op 3.07 (SSa2)

2. Wash with the WP-1 swatches: 3 normal cold cycles in a mesh bag with a towel, line dry

3. Look at every edge with the loupe

Pass: no allowance frayed past the 2nd row on any swatch; a fail on one pairing: hot-knife that cut edge or bind that seam (20 mm grosgrain)

Write in the results sheet (results_template.csv, column measured):

- FRAY-1-01 to FRAY-1-06 (ULTRA|330D to 100D|330D): no fray past the 2nd row after 3 washes. Type Y or N.

Note: BL-038: added; same wash load as WP-1 (WP-1 is the B.2 seared seam)

Defined in: outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv (RD-17); designs/pack_5200/mfg_corrie_f_rev3.py ASSUMPTION NO-BIND; docs/DECISIONS.md 2026-11-13 orchestrator call (3)

2. WP-1: wash-swatch fray, seams without binding

Why: B-1 (no binding on body seams)

Specimen: 4 seam pairings 330D|330D, 330D|ULTRA 200, 500D|330D, 100D|ULTRA; 301 + 2nd row 6 mm in, trimmed, hot-knife seared

Steps

1. Sew one 200 mm swatch per pairing

2. 3 machine-wash cycles (R-12), dry

3. Look at each edge

Pass: no fray past the second row on any swatch

Write in the results sheet (results_template.csv, column measured):

- WP-1-01 to WP-1-04 (330D|330D to 100D|ULTRA): no fray past the 2nd row after 3 washes. Type Y or N.

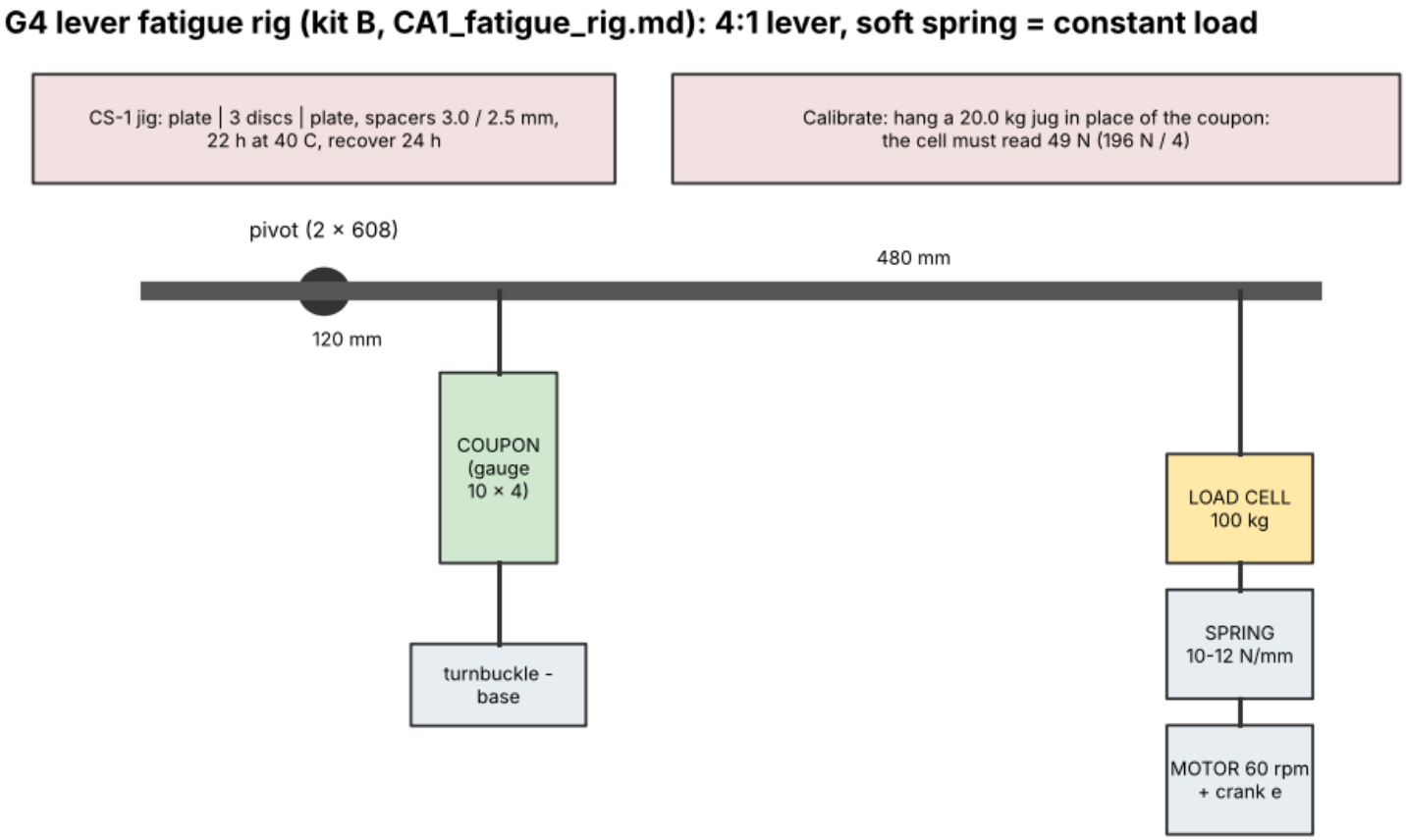
Note: B-1 is already in B.2: a fail adds 36.4 g back. R-12 / S-03 gate is the same test (merged)

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4; designs/pack_5200/analysis/weight_pass_corrie.md B-1; designs/pack_5200/analysis/build_simplicity.md S-03

After the session

1. Fill measured and date for every row you ran; leave the rest empty.
2. Photograph every failed or doubtful sample; put the photo number in photo.
3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

G4: Lever rig: frame coupons (weeks 2-5)



When: build weekend 2, then runs day and night

Rig: kit B lever fatigue rig (CA1_fatigue_rig.md), 40 C box, 3-point bend fixture

Kits: B, D

Build the rig once; every frame coupon runs on it. CA-1 at the release level comes first: frame release R1 cannot happen without it. The 40 C box also runs CS-1 (BL-037) beside CA-4.

About 34.5 h hands-on and 487 h of machine time.

Safety (read before you start)

- The crank and lever move for days: guard the crank with a box, keep hands out while it runs, unplug before touching a coupon.

- 40 C box: thermostat set and checked with a thermometer before you leave it.

- CA-5 static pulls reach 4.4 kN at the stub: blanket over the rig, stand to the side.

Rig sketch

```
            load cell (S-type 100 kg) in the drive link
  motor + crank --> rod --> [spring cage] --> [LC] --> lever end (480 mm from pivot)
                                          |
  pivot (M8 shoulder bolt, 2 x 608) O----------------------+-------- lever 6061 40 x 10 x 600
                                | 120 mm
                          upper grip
                          [coupon]       ratio 480 / 120 = 4
                          lower grip -- turnbuckle -- base
```

CS-1 jig: [plate] disc disc disc [plate] spacers 3.0 mm (6 mm discs) / 2.5 mm (5 mm discs)

4 x M6 bolts, tighten until the plates sit on the spacers, then 22 h in the 40 C box

Picture: sketches/G4_rig.png

Before: the coupons have had their 7 days (G1). Build the rig (one weekend, CA1_fatigue_rig.md 4) and calibrate it with the 20 L jug.

Tests in this session, in order

#	test	releases
1	CA-1	Frame release R1
2	IM3-EC	Frame release R1
3	CA-5	Frame release R1
4	CS-1	BL-037 weight gates
5	BP-1	BL-037 weight gates + Field release and the rest
6	HB-6	Field release and the rest
7	CA-3	Field release and the rest
8	CA-4	Field release and the rest
9	HB-5	Field release and the rest
10	T6	Field release and the rest
11	T7	Field release and the rest
12	CA-2	Field release and the rest

1. CA-1: PAHT-CF fatigue coupons, 0.30 then 0.40

Why: H1 print release R1 (governing flip 0.2556 + 0.02 = 0.276); 0.40 only feeds option F-1 (BL-037)

Specimen: 12 XY + 4 Z + 3 spare bars, printed and annealed like the frame, 7 days at 23 C / ~50 % RH

load	N	lb	kg on the scale
XY bar peak at the release level 0.276	777	175	79.2
XY bar valley (R 0.1)	78	18	8.0
load cell reads (4:1 lever)	194	44	19.8
XY bar peak at 0.40	1126	253	114.8
XY bar peak at 0.404 (only if MT-1c's swing is over 70 %)	1138	256	116.0

Release level today: 0.276 (governing flip 0.2556: body 0.2553, 28 mm IM3 cheek 0.2464, 0.2556 with the pull 10 deg further aft; + 0.02), the level in CA1_fatigue_rig.md section 1 (confirmed by BL-039, analysis/ca1_mt1c_confirm.md). It holds only if MT-1c finds the cinch static (swing <= 0.4 %). Run the 0.40 bars too: they cover a swing up to 70 %. Above that R1 needs 0.404 (0.40 does not cover it).

Steps

1. Calibrate: hang the 20.0 kg jug in place of the coupon; the cell must read 49 N. Write the reading.

2. Static (5 bars): put the turnbuckle in place of the spring; clamp an XY bar 35 mm deep in each grip; turn the turnbuckle about 1 turn every 10 s until the bar breaks; peak cell force x 4 / smallest gauge area = strength in MPa. 3 XY bars, then 2 Z bars (CA1_fatigue_rig.md 6 step 1).

3. UTS_meas = mean of the 3 XY strengths (cap 88 MPa). Work out your own peak: F_max = 0.276 x 0.8 x UTS_meas x area (at 88 MPa and 40 mm2: 777 N; the cell then reads 194 N). F_min = 0.1 x F_max.

4. Set the crank throw and the turnbuckle so the cell swings between F_min / 4 and F_max / 4; check over 20 slow turns by hand before the motor runs.

5. Release level, 3 XY bars one at a time: run to 100,000 cycles (27.8 h at 1 Hz). The rig stops itself if the bar breaks, its stiffness drops more than 10 % or the force drifts more than 5 %.

6. After each run-out look at the gauge and fillets with the loupe; then pull that bar to failure (step 2): it must keep at least 0.9 x UTS_meas.

7. Then 2 Z bars at the release level (0.276 x 0.8 x the Z strength).

8. Only if every release-level bar passed: 3 XY bars at 0.40 (1126 N peak at 88 MPa).

9. 2 of 3 at a level is inconclusive: run 3 more; a level passes only with 5 of 6.

10. Send the rig's CSV log with the results sheet.

Pass: 3/3 run-outs to 1e5 at 0.276, stiffness loss <= 10 %, no crack; static XY >= 79.2 MPa, Z >= 42.3 MPa. R1 uses 0.276 only if MT-1c's swing is <= 0.4 %; up to 70 % it needs the 0.40 pass; above, 3/3 at 0.404 (BL-039)

Write in the results sheet (results_template.csv, column measured):

- CA-1-01 to CA-1-03 (XY#1 to XY#3): static XY UTS. MPa, pass if at least 79.2.

- CA-1-04 to CA-1-05 (Z#1 to Z#2): static Z UTS. MPa, pass if at least 42.3.

- CA-1-06 to CA-1-08 (RXY#1 to RXY#3): cycles at the release level 0.276 (run-out 100000). Cycles, pass if at least 100000.

- CA-1-09 to CA-1-11 (SXY#1 to SXY#3): stiffness loss at run-out. %, pass if at most 10.

- CA-1-12 to CA-1-13 (RZ#1 to RZ#2): Z cycles at 0.276. Cycles, pass if at least 100000.

- CA-1@0.40-01 to CA-1@0.40-03 (XY#1 to XY#3): cycles at 0.40 (run-out 100000). Cycles, pass if at least 100000.

Note: BL-038: release level from h1_buckling.json (BL-027 body flip) and im3_body_simloads.json (28 mm cheek), replacing the 0.310 placeholder (CA1_fatigue_rig.md 1 asked for this). Grams: BL-037 F-1 (-9.9 g, an option that also needs a buckling re-run), not -50..-70 g. Machine: 8 runs x 27.8 h. CT-B1 merged here

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md; designs/pack_5200/analysis/frame_comp_verdict.md §6

2. IM3-EC: IM3 eye cycle coupons, 3 x 1e5 at >= 1.15 x 1024 N

Why: H1 print release (IM3 eye); P_cheek

Specimen: 3 eye coupons: 120 mm stay segment round z 520 with the IM3 eye and the 28 mm cheek (BL-021), cut from the H1 STL, printed with the CA-1 bars

load	N	lb	kg on the scale
eye cycle peak (1.15 x 1024 N)	1178	265	120.1
after run-out, proof pull	2553	574	260.2

Steps

1. Sewn T3 tongue + IM3 carabiner through the eye, led 22 deg aft of the stay axis, coupon clamped by the stay ends

2. Lever rig at R 0.1: 118 -> 1178 N, 1 Hz, to 1e5

3. Loupe check every 25,000

4. Then pull to 2553 N on the pull bench

Pass: 3/3 run-outs to 1e5 at 1178 N, no crack at 10x; residual pull 2553 N with no crack

Write in the results sheet (results_template.csv, column measured):

- IM3-EC-01 to IM3-EC-03 (#1 to #3): cycles at 118 -> 1178 N (run-out 100000). Cycles, pass if at least 100000.

- IM3-EC-04 to IM3-EC-06 (p#1 to p#3): no crack at 10x and held 2553 N after. Type Y or N.

Note: BL-038: cheek 27 -> 28 mm (BL-021 adopted). BL-039: load 1178 N from the BL-021 sim demand (was 1017 N = 1.15 x 1.6 x 552.4 N, BL-010's 15 in IM3 force, below today's demand)

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §6a; designs/pack_5200/analysis/im3_body_simloads.md (28 mm cheek, governing hook + strap); designs/pack_5200/analysis/loadpath_verification.md §4.1, §8

3. CA-5: dry z 234 joint stub pulls + 1e5 cycle

Why: H1 print release

Specimen: 4 stub pairs, one stay, 60 mm each side of z 234, real bosses, Ti bolts

load	N	lb	kg on the scale
static mean must reach	4400	989	448.5
lowest static	3666	824	373.7
cycle force	1543	347	157.3

Steps

1. 3 static eccentric pulls to failure (200 kg cell or 6:1 hole)

2. 1 stub 1e5 cycles at the 1.6 g force with a dial across the joint

Pass: static mean >= 4400 N, lowest >= 3666 N (1.5 / 1.25 x 2933 N); cycle: no crack at the boss root, joint step <= 0.1 mm

Write in the results sheet (results_template.csv, column measured):

- CA-5-01 to CA-5-03 (#1 to #3): static stub peak. N, pass if at least 3666.

- CA-5-04 (mean): mean of the 3 static peaks. N, pass if at least 4400.

- CA-5-05 (cyc): joint step after 1e5 cycles at 1543 N. Mm, pass if at most 0.1.

- CA-5-06 (cyc): no crack at the boss root after 1e5 cycles. Type Y or N.

Note: CONFLICT: verdict uses 2894 N and 0 -> 1543 N (team A); CA1 doc uses 2933 N (H1). 2933 N used. H1 cycle force not restated: 1543 N used until structural gives the H1 value. Fail -> B 8-bolt flange, +17 to +33 g. CT-B3 merged

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §7; designs/pack_5200/analysis/frame_h1.md §8; designs/pack_5200/analysis/frame_comp_verdict.md §6

4. CS-1: XLPE L200 compression set, 50 % for 22 h at 40 C

Why: BL-037 B-1 (-31.9 g) and H-1 (-22.3 g)

Specimen: 3 discs 30 mm diameter of XLPE L200 6 mm (belt) and 3 of 5 mm (straps)

Steps

1. Measure each disc's thickness t0 with the caliper (light touch, 3 points, mean)

2. Put the discs between the two jig plates with 3.0 mm spacers (6 mm discs) or 2.5 mm spacers (5 mm discs)

3. Tighten until the plates sit on the spacers

4. 22 h in the 40 C box

5. Take out, unclamp, let them recover 24 h on a flat table

6. Measure tf the same way

7. Set % = (t0 - tf) / t0 x 100

Pass: every disc set <= 15 % (the EVA 45C record value)

Write in the results sheet (results_template.csv, column measured):

- CS-1-01 to CS-1-03 (6mm#1 to 6mm#3): compression set, 6 mm disc. %, pass if at most 15.

- CS-1-04 to CS-1-06 (5mm#1 to 5mm#3): compression set, 5 mm disc. %, pass if at most 15.

Note: BL-038: added (BL-037). Shares the CA-4 40 C box (ASTM D3575 style)

Defined in: designs/pack_5200/analysis/weight_push.md 2 (B-1, H-1), 7 XLPE-SET; data/materials/foams/xlpe_foam_l200.yaml (12 %), eva_foam_45c.yaml (15 %)

5. BP-1: sandwich backer strip 3-point bend coupon

Why: BL-005 sandwich strip (A_back); BL-037 K-1 (2 mm ROHACELL 71 core)

Specimen: 70 x 200 mm offcuts of the strip layup: 3 mm and 2 mm ROHACELL 71 core (2 mm: sand a 3 mm sheet or special order)

Steps

1. 100 mm span on two 10 mm rods, load at mid-span with a 10 mm rod

2. Dead weight 3.06 kg (30 N)

3. Dial indicator at mid-span

4. 3 loadings

Pass: mean mid-span deflection 0.205-0.277 mm (model 0.241 mm +/- 15 %)

Write in the results sheet (results_template.csv, column measured):

- BP-1-01 (3mm): 3 mm core strip passes BP-1. Type Y or N.

- BP-1-02 (2mm): 2 mm core strip passes BP-1 (SF >= 1 at the factored strip load). Type Y or N.

Note: model 0.241 mm = 0.134 bending (D11 66,647 N mm) + 0.107 core shear (S 100.2 N/mm), per mm width x 70 mm, back_system2 RECOMMENDED (computed by BL-023). Fail: BP-08 twill backer back (+44.6 g)

Defined in: designs/pack_5200/analysis/back_system_integration.md §6

6. HB-6: M5 heat-set insert pull-out in printed PAHT-CF

Why: hip pivot mount on H1

Specimen: 5 inserts in printed, annealed PAHT-CF blocks

Steps

1. Bolt an M5 eye into each insert

2. Pull along the insert axis to 1.5 kN

Pass: >= 1.5 kN, 5 of 5

Write in the results sheet (results_template.csv, column measured):

- HB-6-01 (1): passes the register criterion: >= 1.5 kN, 5 of 5. Type Y or N.

Note: D-I1 (frame D) and CT-B5 (frame B) merged here

Defined in: designs/pack_5200/analysis/hipbelt_structural.md §8; designs/pack_5200/analysis/fabric_swap_evidence.md (release gate)

7. CA-3: interlayer after 40 C water soak

Why: FI interlayer term; plug shear-out

Specimen: 3 Z bars + 3 short-beam blocks 24 x 8 x 6

Steps

1. 48 h in 40 C water

2. Static Z pull

3. 3-point short-beam shear, span 5 x thickness

Pass: Z >= 37.6 MPa; interlayer shear >= 22.6 MPa

Write in the results sheet (results_template.csv, column measured):

- CA-3-01 to CA-3-03 (Z#1 to Z#3): Z strength after 48 h at 40 C water. MPa, pass if at least 37.6.

- CA-3-04 to CA-3-06 (SB#1 to SB#3): short-beam interlayer shear. MPa, pass if at least 22.6.

Note: CT-B2 merged

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §7; designs/pack_5200/analysis/frame_comp_A.md

8. CA-4: creep-rupture 40 C 8 h

Why: creep margin

Specimen: 2 XY bars

load	N	lb	kg on the scale
dead load at the lever end (4:1)	282	63	28.7

Steps

1. Dead weight 29 kg on the lever end (282 N link)

2. Bars in a 40 C box for 8 h

3. Log stretch

Pass: no rupture

Write in the results sheet (results_template.csv, column measured):

- CA-4-01 to CA-4-02 (#1 to #2): no rupture after 8 h at 40 C (282 N link). Type Y or N.

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §7

9. HB-5: 7075-T6 1.2 bow-tie plate proof + 1e5 cycles

Why: C-1 + C-2 + C-3

Specimen: 1 plate on the Ti pin over a foam block

Steps

1. Template-check end slope 14.6 +/- 2 deg

2. 1.0 kN per ear + 1.6 kN vertical with a scribed grid

3. Then 1e5 walking cycles

4. Re-check slope

Pass: no yield (grid unchanged), slope within tolerance after cycling, no crack

Write in the results sheet (results_template.csv, column measured):

- HB-5-01 (1): passes the register criterion: no yield (grid unchanged), slope within tolerance after cycling, no crack. Type Y or N.

Note: CONFLICT: two definitions (6061 1.5 vs 7075 1.2); 7075 adopted 2026-11-09

Defined in: designs/pack_5200/analysis/weight_c_structural.md §7 (supersedes designs/pack_5200/analysis/hipbelt_structural.md §8 6061 version)

10. T6: crown-corner bond coupons (0.8 mm bondline)

Why: carbon crown corners + extension-post socket (P-1)

Specimen: 3 corner coupons, 0.8 mm glass-bead bondline

Steps

1. Build like the frame

2. Cure

3. Pull to failure

Pass: mean >= 1.5 x, lowest >= 1.25 x the largest FEA limit end force (frame_sizing_cases_19in.md); cohesive failure

Write in the results sheet (results_template.csv, column measured):

- T6-01 (1): passes the register criterion: mean >= 1.5 x, lowest >= 1.25 x the largest FEA limit end force (frame_sizing_cases_19in.md); cohesive failure. Type Y or N.

Note: type (b) collar-on-stay retired: H1 stays are printed. Numeric pass force not restated in any test doc (FLAG)

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/print/frame/PRODUCTION.md (T6 note)

11. T7: bushing crush coupon

Why: crown bushing ring crush credit

Specimen: 150 mm printed H1 sleeve segment + UHMWPE bushing + leg offcut

Steps

1. Press the leg sideways at the bushing centre to 1.5 x the max FEA bushing force, unload, measure

2. Then to failure

Pass: no crack; ovalisation <= 0.2 mm after unload at 1.5 x

Write in the results sheet (results_template.csv, column measured):

- T7-01 (1): passes the register criterion: no crack; ovalisation <= 0.2 mm after unload at 1.5 x. Type Y or N.

Note: CONFLICT: defined on a CFRP stay with a wrap collar; H1 carries the bushing in a printed sleeve, so the specimen is revised. Force value not restated (FLAG)

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

12. CA-2: anneal 90 C vs 120 C

Why: nothing until a gain is measured

Specimen: 3 XY + 3 Z bars per anneal

Steps

1. Static pulls of both sets

Pass: credit only a measured gain (0-11 % expected)

Write in the results sheet (results_template.csv, column measured):

- CA-2-01 (1): passes the register criterion: credit only a measured gain (0-11 % expected). Type Y or N.

Note: grams not quantified anywhere

Defined in: designs/pack_5200/analysis/frame_comp_A.md; docs/research/frame_comp_A.md

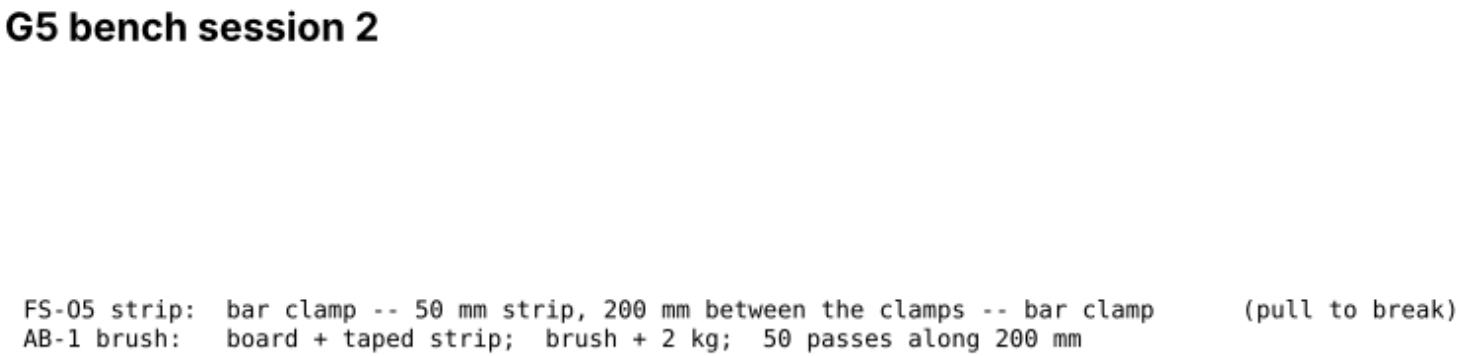
After the session

1. Fill measured and date for every row you ran; leave the rest empty.

2. Photograph every failed or doubtful sample; put the photo number in photo.

3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

G5: Bench session 2: fabric, belt and back coupons



When: any weekend while the lever rig runs (weeks 2-4)

Rig: kit A pull bench + kit D

Kits: A, D

Sewn and laminated coupons that do not need the frame. Send the AB-1 lab samples on the first day: the lab takes about 4 weeks.

About 9.9 h hands-on and 8 h of machine time.

Safety (read before you start)

- Same bench rules as G2. FS-O5 strips break suddenly: blanket over the clamps.

Rig sketch

FS-O5 strip: bar clamp -- 50 mm strip, 200 mm between the clamps -- bar clamp (pull to break)

AB-1 brush: board + taped strip; brush + 2 kg; 50 passes along 200 mm

Picture: sketches/G5_rig.png

Make first: the specimens listed under each test (register specimen). Ship the AB-1 lab samples on day 1.

Tests in this session, in order

#	test	releases
1	WP-2	Bag release, step 2: batch
2	AB-1	BL-037 weight gates
3	HB-4	Field release and the rest
4	HB-T	Field release and the rest
5	PS-1	Field release and the rest
6	BP-2	Field release and the rest
7	HB-2	Field release and the rest

1. WP-2: TPU 840D lash-tab slot pull (C-6 / O-15)

Why: C-6 lash tabs Hypalon -> TPU 840D

Specimen: 5 laser-cut B-11 tabs

Steps

1. Strap through the slot pulled in its working direction to 450 N

2. Hold 60 s

Pass: 5 of 5 hold 450 N, no slot tear

Write in the results sheet (results_template.csv, column measured):

- WP-2-01 (1): passes the register criterion: 5 of 5 hold 450 N, no slot tear. Type Y or N.

Note: slot direction still to be confirmed on the tab drawing (materials-engineer flag)

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4; designs/pack_5200/analysis/weight_pass_corrie.md C-6

2. AB-1: HyperD 300 vs 630D printable: Martindale 8000 rev, brush drag, print swatch

Why: BL-037 G-2 (-22.5 g)

Specimen: 1 yd HyperD 300 printable with the owner's camo (custom print); a 630D printable offcut; 2 lab samples 300 x 300 mm

Steps

1. Order the HyperD print (an 8 x 8 in swatch first if offered)

2. Send one 300 x 300 mm sample of each fabric to a textile lab for Martindale abrasion (ASTM D4966 or ISO 12947-2) to 8000 rev, end point a hole or 2 broken yarns

3. At home tape a 50 x 300 mm strip of each to a board and drag a stiff nylon brush loaded with 2 kg along 200 mm, 50 passes

4. Look with the loupe

5. Judge the printed camo in daylight at 1 m and 10 m

Pass: HyperD: no hole at 8000 rev; brush strip no worse than the 630D strip; owner OK on the print in writing

Write in the results sheet (results_template.csv, column measured):

- AB-1-01 (lab): HyperD 300: no hole at 8000 rev. Type Y or N.

- AB-1-02 (lab): 630D: rev at end point (reference). Type the value in rev.

- AB-1-03 (brush): HyperD strip no worse than 630D after 50 passes. Type Y or N.

- AB-1-04 (print): owner OK on the camo print (in writing). Type Y or N.

Note: BL-038: added (BL-037). LONG LEAD: lab turnaround about 4 weeks after receipt; custom print lead time unknown

Defined in: designs/pack_5200/analysis/weight_push.md 2 (G-2), 7 HYPERD-ABRASION; data/materials/textiles/polyester_300d_printable.yaml

3. HB-4: hipbelt stitched-ear joint

Why: belt B wing joint

Specimen: 1 ear stitched into a wing-root coupon (3 recommended)

Steps

1. Clamp the wing coupon

2. Pull the ear at 20 deg to 1.5 kN

3. Measure the holes

Pass: no thread break; hole elongation < 0.5 mm

Write in the results sheet (results_template.csv, column measured):

- HB-4-01 (1): passes the register criterion: no thread break; hole elongation < 0.5 mm. Type Y or N.

Note: defined on the 6061 1.5 mm plate; repeat on the adopted 7075 1.2 mm plate

Defined in: designs/pack_5200/analysis/hipbelt_structural.md §8

4. HB-T: 50 mm buckle tab + 25 mm anchor and lock-tab box-X pull

Why: belt B 50 mm buckle tabs (C8 booking)

Specimen: 5 tabs sewn like the belt: 50 mm webbing round a buckle bar, 25 mm anchor and lock-tab box-X

Steps

1. Tab round a steel bar (no buckle) on the pin fixture

2. Pull with the come-along to 1500 N (152.9 kg)

3. Hold 60 s

Pass: 5 of 5 hold 1500 N for 60 s, no broken stitch

Write in the results sheet (results_template.csv, column measured):

- HB-T-01 (1): passes the register criterion: 5 of 5 hold 1500 N for 60 s, no broken stitch. Type Y or N.

Note: 1500 N is above the 1480 N ratchet-strap limit: come-along only

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 H1, §8

5. PS-1: pad-seat backer sag

Why: C-4 correction (pad seat)

Specimen: pad seat on a printed pan offcut

Steps

1. 413 N (42.1 kg) on a 150 x 190 foam block for 8 h

2. Then 661 N (67.4 kg)

3. Dial in the channel

Pass: sag < 3 mm; no contact with a dummy plate spine at 4.5 mm

Write in the results sheet (results_template.csv, column measured):

- PS-1-01 (1): passes the register criterion: sag < 3 mm; no contact with a dummy plate spine at 4.5 mm. Type Y or N.

Note: CONFLICT: corrie_body has 413 N only and 'no contact'; weight_c adds 661 N and < 3 mm. Stricter weight_c version used

Defined in: designs/pack_5200/analysis/weight_c_structural.md §7; designs/pack_5200/analysis/corrie_body.md §12.4

6. BP-2: stay contact width on the backer strip (pressure film)

Why: BL-005 sandwich strip (A_back)

Specimen: 100 mm printed H1 stay segment (z about 260) on a strip offcut, 10 mm foam under the strip

Steps

1. Carbon paper + white paper (or a Prescale LW sheet) between stay and strip

2. Dead load 25.5 kg (2.5 N/mm x 100 mm) for 2 min

3. Measure the imprint width at 5 stations

Pass: imprint width >= 8 mm at all 5 stations

Write in the results sheet (results_template.csv, column measured):

- BP-2-01 (1): passes the register criterion: imprint width >= 8 mm at all 5 stations. Type Y or N.

Note: ASSUMPTION: a carbon-paper imprint stands in for pressure film; if the width is within 2 mm of 8 mm, buy one Prescale LW sheet and repeat. Fail: structural re-checks core crush on the measured width

Defined in: designs/pack_5200/analysis/back_system_integration.md §6

7. HB-2: belt face friction

Why: silicone grip print (A-1; OD-HB4)

Specimen: belt face fabric with and without the grip print; polyester and merino base layers, dry and sweaty

Steps

1. Base layer taped to a board

2. 500 g weighted sled with the face fabric on it

3. Raise one end slowly until it slides

4. Read the angle in phyphox

5. Mu = tan(angle)

6. 5 runs each

Pass: mu >= 0.45 credits the grip print

Write in the results sheet (results_template.csv, column measured):

- HB-2-01 (1): passes the register criterion: mu >= 0.45 credits the grip print. Type Y or N.

Note: ASSUMPTION: tilt-board static mu replaces the ASTM D1894-style sled pull (no low-range force gauge in the kits); grip print adds 4.9 g if adopted

Defined in: designs/pack_5200/analysis/hipbelt_fit.md §10

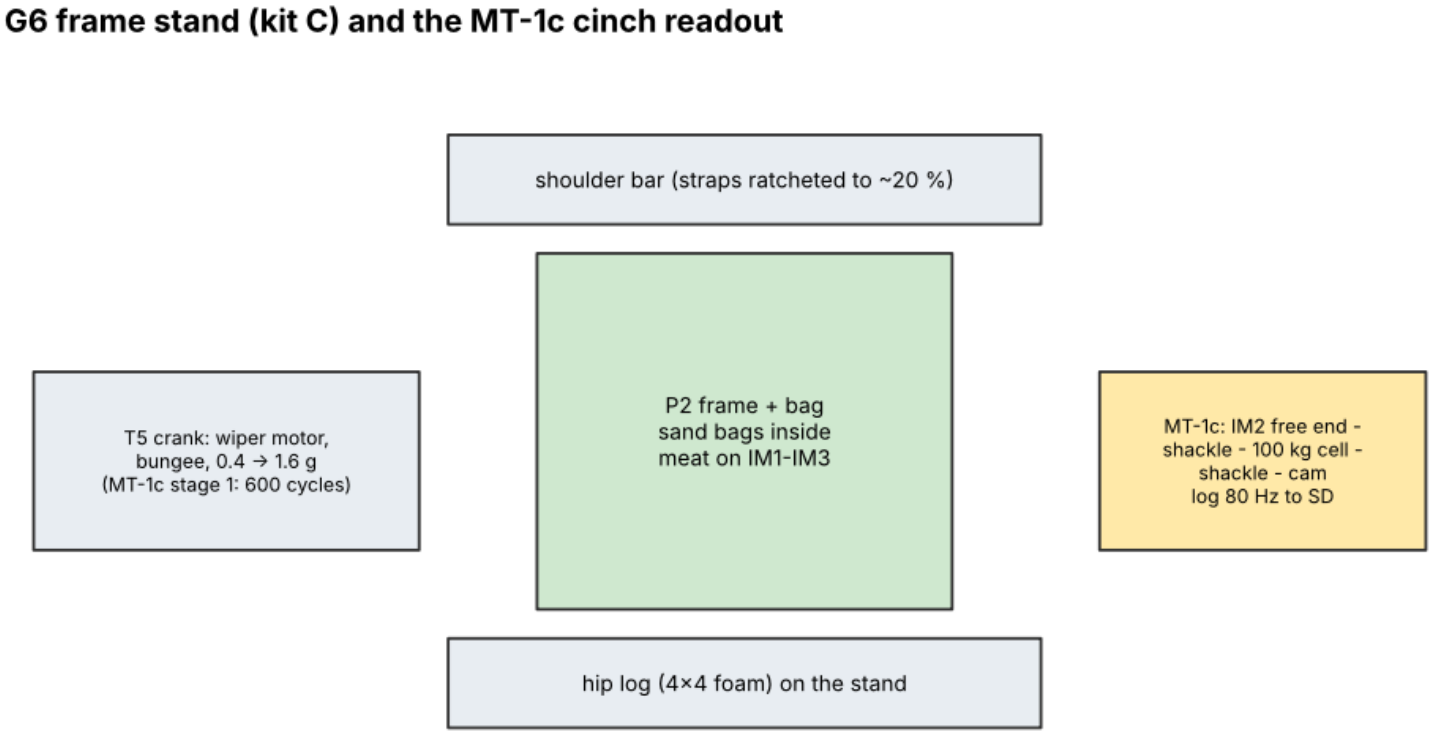
After the session

1. Fill measured and date for every row you ran; leave the rest empty.

2. Photograph every failed or doubtful sample; put the photo number in photo.

3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

G6: First-article frames on the stand (P1 then P2)



When: after G4's CA-1, IM3-EC and CA-5 pass (about week 5-6)

Rig: kit C frame test stand (hip log, shoulder bar, sand bags, T5 crank) + kit A scale and come-along

Kits: C, A, B (load cell), D

Frame release R1 completes here: IM3-EP on P1, then T9, T1, T2 and MT-1c on P2. Nobody carries the frame until T1 to T4 have passed. T2u destroys P1 and is always last.

About 52.2 h hands-on and 37 h of machine time.

Safety (read before you start)

- Back-up sling under every load. Nobody under the load. T2u (P1, 204 kg) outdoors with a 3 m exclusion zone.
- Nobody carries the frame until T1, T3, T2 and T4 have passed. MT-1c stage 1 runs on the stand for that reason.

- T4c: -20 C parts are brittle; gloves and glasses.

Rig sketch

crown top bar --- shoulder bar (straps ratcheted to carry ~20 %)

|

[P2 frame + bag, sand bags inside]

|

hip log (foam 4x4) on the stand come-along + crane scale pull down for T1 / T2 / T3

|

T5 crank: wiper motor -- crank -- bungee -- bag (0.4 -> 1.6 g)

MT-1c: IM2 strap free end -- [snap shackle] -- [100 kg S cell] -- [snap shackle] -- cam

HX711 + Nano + SD logger taped to the frame, 80 Hz

Picture: sketches/G6_rig.png

Before: CA-1 at the release level, IM3-EC and CA-5 have passed (G4), and MT-1 (G2). Print P1 and P2 from the released files, same spool and profile as the coupons. Sew the bag (G2 + G3 passed).

Tests in this session, in order

#	test	releases
1	IM3-EP	Frame release R1
2	W-PULL	Frame release R1
3	T9	Frame release R1 + Bag release, step 2: batch
4	T9-R3a	Bag release, step 2: batch
5	T9-R3c	Bag release, step 2: batch
6	BP-3	Field release and the rest
7	RF-1	Field release and the rest
8	T1	Frame release R1
9	T3	Field release and the rest
10	T2	Frame release R1
11	MT-1c	Frame release R1
12	CR-1	Field release and the rest
13	BU-1	BL-037 weight gates
14	SP-1	Field release and the rest
15	T4a	Field release and the rest
16	T4b	Field release and the rest
17	T4c	Field release and the rest

#	test	releases
18	T5	Field release and the rest
19	T5f	Field release and the rest
20	T2u	Field release and the rest

1. IM3-EP: IM3 eye pull on the first printed frame

Why: H1 release; P_cheek

Specimen: P1 frame (the sacrificial frame), both IM3 eyes

load	N	lb	kg on the scale
eye pull	2553	574	260.2

Steps

1. Frame clamped to the kit C stand

2. Sewn T3 tongue + carabiner through the eye

3. Pull along the rev C IM3 vector (frame_h1_results.json) to 2553 N (260.2 kg)

4. Hold 60 s

5. Loupe

Pass: both eyes hold 2553 N for 60 s, no crack, no whitening at the cheek root

Write in the results sheet (results_template.csv, column measured):

- IM3-EP-01 (1): passes the register criterion: both eyes hold 2553 N for 60 s, no crack, no whitening at the cheek root. Type Y or N.

Note: on P1 before T2u; P2 is not pulled. The throat fit check (two carabiner rods) is done here too

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8 (FAIRLEAD_DIR), §4.1

2. W-PULL: W1-W3 lash points on the first frame

Why: W1-W3 frame anchors (BL-003)

Specimen: P1 frame, 6 lash points (z 80 / 250 / 420, both sides)

Steps

1. 25 mm webbing loop through each lash point

2. Pull along the strap line to 338 N (34.5 kg)

3. Hold 60 s

Pass: 6 of 6 hold 338 N, no crack, no set

Write in the results sheet (results_template.csv, column measured):

- W-PULL-01 (1): passes the register criterion: 6 of 6 hold 338 N, no crack, no set. Type Y or N.

Note: same session as IM3-EP

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 S8, S9, §8

3. T9: loaded CG (+ 15 in, 30 kg meat check); CORRIE F rev 3 items in rows T9-R3a..d (BL-038 split)

Why: HA-LS-1 rev C basis; before T1; CORRIE F rev 3 (LID-SIM, CROWN_PRE_N, ROLL-PATH, LID-SEAT, LID_SLIDER5_G, LID_THREAD)

Specimen: P2 frame + bag (rev C CG at 15 / 19 / 23 in); the rev 3 items are rows T9-R3a..d

load	N	lb	kg on the scale
fill	667	150	68.0
meat check	294	66	30.0

Steps

1. 68 kg fill and meat on IM1-IM3 at 15/19/23 in

2. Hang from haul loop, then from a hip pin, plumb both lines

3. Add 30 kg on IM1-IM3 at 15 in

4. Then rows T9-R3a (port guys, S kit CG), T9-R3c (crown strap path) in the same session

5. T9-R3b (cinch after a walk) and T9-R3d (lid) after T1-T4

Pass: bag CG <= 205 mm at z <= 590; meat <= 190; tall <= 185; rev 3 items: rows T9-R3a..d

Write in the results sheet (results_template.csv, column measured):

- T9-01 to T9-03 (15in to 23in): meat CG aft of the stay plane. Mm, pass if at most 190.

- T9-04 (fill): bag CG aft (uniform fill). Mm, pass if at most 205.

- T9-05 (tall): tall-load CG aft. Mm, pass if at most 185.

Note: corrie 11.4 plumb check and MOD-5 merged here; BL-038 split the rev 3 items (BL-030) into T9-R3a..d and moved the crown straps to BL-031's 25 mm per-kit cuts

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/corrie_body.md §11.4; designs/pack_5200/analysis/corrie_f_body.md §14.10, §15.6 (BL-030)

4. T9-R3a: T9 rev 3 (a): port guys hold the S-kit top, meat CG at 15 in

Why: CORRIE F rev 3 top as drawn (bag batch release)

Specimen: P2 frame, S crown kit (15 in), sewn rev 3 bag, port guys fitted, lid on with 1 kg inside

load	N	lb	kg on the scale
fill incl. 30 kg meat	667	150	68.0

Steps

1. Top packed full above the guys

2. 30 kg sand meat on IM1-IM3, fill to 68 kg

3. Hang from the haul loop and plumb a line

4. Hang from a pin through the hip pivot and plumb a second line

5. The CG is where the lines cross

6. Measure its distance aft of the stay plane

Pass: meat CG <= 190 mm aft (record against 179.3 mm)

Write in the results sheet (results_template.csv, column measured):

- T9-R3a-01 (S15): meat CG aft, S kit, top full, lid + 1 kg (sim 179.3). Mm, pass if at most 190.

Note: BL-038: split out of T9 (was T9 rev 3 item a); sim value updated to BL-033's 179.3 mm

Defined in: designs/pack_5200/analysis/corrie_f_body.md 14.5, 14.10; docs/DECISIONS.md 2026-11-13 BL-033 row (179.3 mm)

5. T9-R3c: T9 rev 3 (c): crown strap free length (ROLL-PATH, TORSO-LIN)

Why: B-28 cut per crown kit (bag batch release)

Specimen: sewn rev 3 bag on P2 with the S kit at its short end (15 in) packed full, collar standing; then the L kit at 21.5 in, 6-day load

Steps

1. Cinch both crown straps

2. Measure the strap path from the bar tab to the ladder lock along the strap with a tape (both straps)

3. Repeat at the L kit 6-day setting

Pass: S kit full: path <= 968.1 mm (cut 1068.1 - 2 x 50 mm grip tail; sim 928.1 mm); L kit 6-day: record against the sim's 438.8 mm

Write in the results sheet (results_template.csv, column measured):

- T9-R3c-01 to T9-R3c-02 (S-full-L to S-full-R): crown strap path, S kit full. Mm, pass if at most 968.1.

- T9-R3c-03 (L-6day): crown strap path, L kit 6-day (sim 438.8). Type the value in mm.

Note: BL-038: split out of T9 (item c). BL-030's 1138 mm single 20 mm cut is replaced by BL-031's per-kit 25 mm cuts. BL-039 added 40 mm spare to every kit (BL-031 had none: the S sim path 928.1 mm equalled its limit); a path above 968.1 mm still means a longer S cut

Defined in: designs/pack_5200/analysis/corrie_f_body.md 15.6, 16.2, 16.5

6. BP-3: back pressure on the rails, below and above z 190

Why: back system release; BL-005 (A_back) skin model

Specimen: P2 frame + bag + Sandwich v2 back, crown kits S / M / L (15 / 19 / 23 in), 3 testers

Steps

1. Calibrate 8 FSRs on a 12.7 mm puck with 0.5 / 1 / 2 kg

2. Tape 4 per side on the stay line at z 100, 160 (below 190) and z 260, 400

3. T9 loaded state, 68 kg meat on IM1-IM3

4. Stand 2 min and walk 5 min at each torso

5. Read peaks

Pass: every sensor <= 14 kPa at 15, 19 and 23 in; record whether the z 100 / 160 sensors read > 1 kPa (rails bear below z 190)

Write in the results sheet (results_template.csv, column measured):

- BP-3-01 (1): passes the register criterion: every sensor <= 14 kPa at 15, 19 and 23 in; record whether the z 100 / 160 sensors read > 1 kPa (rails bear below z 190). Type Y or N.

Note: run in the T9 session with RF-1. Fail: lumbar-pad shim (structural to size; ASSUMPTION fallback, mass not yet booked) and re-test. FSR scatter about 10 % (ASSUMPTION): a reading within 1.4 kPa of the limit is repeated

Defined in: designs/pack_5200/analysis/back_system_integration.md §6, §8; designs/pack_5200/analysis/review_20261111.md S4

7. RF-1: back rail fit 130/70 on 3 backs (C-8)

Why: C-8

Specimen: rails on P2, 3 testers

Steps

1. FSRs under the rails at the stay line

2. 68 kg

3. 3 testers

Pass: peak <= 14 kPa and comfort OK

Write in the results sheet (results_template.csv, column measured):

- RF-1-01 (1): passes the register criterion: peak <= 14 kPa and comfort OK. Type Y or N.

Note: run with BP-3 (same FSR set-up, same session)

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4; designs/pack_5200/analysis/weight_pass_corrie.md C-8

8. T1: static proof 150 / 300 / 225 lb (+ joint marks, yoke dial)

Why: first loaded carry

Specimen: P2 frame + bag

load	N	lb	kg on the scale
(a)	667	150	68.0
(b)	1334	300	136.0
(c) hipbelt off	1001	225	102.0

Steps

1. (a) 150 lb, (b) 300 lb, (c) belt off 225 lb 2 min

2. Dial on crown, lumbar and the yoke tip

3. Pencil marks across z 234

Pass: no crack, tap clean, crown set <= 1 mm, crown slip 0, re-load within 10 %, FEA +/-30 %; z 234 step <= 0.5 mm after unload

Write in the results sheet (results_template.csv, column measured):

- T1-01 (1): passes the register criterion: no crack, tap clean, crown set <= 1 mm, crown slip 0, re-load within 10 %, FEA +/-30 %; z 234 step <= 0.5 mm after unload. Type Y or N.

Note: D-T1 torsion check is referenced in frame_h1.md §8 but never defined (FLAG)

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/frame_h1.md §8; designs/pack_5200/analysis/weight_c_structural.md §7

9. T3: side-hill 240 lb at 25 deg

Why: first loaded carry

Specimen: P2 frame

load	N	lb	kg on the scale
side-hill 25 deg	1068	240	108.8

Steps

1. Tilt rig 25 deg, downhill strap at 25 % tension, 240 lb pushed 30 % downhill

Pass: as T1; crown legs slide freely after

Write in the results sheet (results_template.csv, column measured):

- T3-01 (1): passes the register criterion: as T1; crown legs slide freely after. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

10. T2: meat / tall-load proof 300 lb

Why: first meat haul

Specimen: P2 frame + bag

load	N	lb	kg on the scale
meat / tall load	1334	300	136.0

Steps

1. 300 lb sandbags in game bags inside the bag on IM1-IM3

2. Tall variant lashed to crown bar

Pass: as T1

Write in the results sheet (results_template.csv, column measured):

- T2-01 (1): passes the register criterion: as T1. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

11. MT-1c: meat-strap cinch readout: does the IM cinch cycle with the walking g (CINCH_STATIC)

Why: frame release R1: body fatigue 1.175 with the cinch static, 0.782 if it cycles fully; the measured swing sets the CA-1 level R1 needs (BL-039, designs/pack_5200/analysis/ca1_mt1c_confirm.md)

Specimen: P2 frame + sewn bag, 30 kg sand meat on IM1-IM3 cinched by hand as in the field; kit B 100 kg S-type cell + HX711 inline in the IM2 strap's free end (two snap shackles), logging to an SD card

load	N	lb	kg on the scale
meat on IM1-IM3	294	66	30.0
pack fill	667	150	68.0

Steps

1. Cinch IM1-IM3 by hand as in the field and lock the cams

2. Log the IM2 strap tension at 80 Hz

3. Stage 1 on the stand: run the T5 crank 600 cycles (10 min, 0.4 -> 1.6 g) with the meat in

4. Stage 2 after T1-T4 pass: a 10 min loaded walk on rough ground

5. For each stage compute the mean tension and the mean peak-to-peak swing per cycle / step

6. Send the log

Pass: record the larger peak-to-peak swing / mean of the two stages (%). R1 needs: swing <= 0.4 %: CA-1 at 0.276; swing <= 70 %: CA-1 also passed at 0.40; above: CA-1 at 0.404 (BL-039 table, ca1_mt1c_confirm.md section 2; replaces BL-038's 10 % pass bound, which keeps margin >= 1.0 only up to 11.6 % and without the 0.02 cushion)

Write in the results sheet (results_template.csv, column measured):

- MT-1c-01 (stand): mean cinch tension, T5 crank 600 cycles. Type the value in N.

- MT-1c-02 (stand): peak-to-peak swing / mean tension (sets the CA-1 level, ca1_mt1c_confirm.md). Type the value in %.

- MT-1c-03 (walk): mean cinch tension, 10 min walk. Type the value in N.

- MT-1c-04 (walk): peak-to-peak swing / mean tension (sets the CA-1 level, ca1_mt1c_confirm.md). Type the value in %.

Note: BL-038: added. The register's MT-1 is the box-X proof; the R1 hold named 'MT-1' in DECISIONS is this cinch readout (h1_buckling.md). Stage 1 replaces the walk for R1 (no person under an unreleased frame); stage 2 confirms

Defined in: designs/pack_5200/analysis/h1_buckling.md (CINCH_STATIC); designs/pack_5200/analysis/h1_sleeve.md; docs/DECISIONS.md 2026-11-12 BL-024 / BL-027 rows (R1 waits on MT-1)

12. CR-1: crown retention hang 136 kg

Why: torso adjust

Specimen: P2 frame

load	N	lb	kg on the scale
crown hang	1334	300	136.0

Steps

1. Run the crown through full travel (pawl clicks)

2. Hang 136 kg from the crown top bar, frame upside down then upright

Pass: no slip either way

Write in the results sheet (results_template.csv, column measured):

- CR-1-01 (1): passes the register criterion: no slip either way. Type Y or N.

Defined in: designs/pack_5200/BUILD_PLAN.md §5

13. BU-1: 50 mm crown bushings: kit change and pawl travel, gloved

Why: BL-037 F-3 (-4.0 g)

Specimen: 2 UHMWPE crown bushings cut 50 mm from the 70 mm stock, on P2

Steps

1. Fit the 50 mm bushings

2. Winter gloves on

3. Change the crown kit S -> M -> L -> S and run the torso pawl through every stop, 20 cycles

4. Note any stick

Pass: 20 of 20 without binding (slides by hand, no tool); the pawl engages every stop

Write in the results sheet (results_template.csv, column measured):

- BU-1-01 (1): gloved kit / pawl cycles without binding, out of 20. Count, pass if at least 20.

Note: BL-038: added (BL-037); CR-1 session

Defined in: designs/pack_5200/analysis/weight_push.md 2 (F-3), 7 BUSHING-LD; designs/pack_5200/analysis/crown_diet (e)

14. SP-1: spreader bone-end proof 400 N

Why: spreader release

Specimen: P2 spreader

load	N	lb	kg on the scale
bone-end proof	400	90	40.8

Steps

1. 400 N through a 22 mm dowel at window centre, US2 corner relief, IM1 notch root

2. Coin tap before/after

Pass: no crack, no skin-core disbond, deflection inside the 14 mm standoff

Write in the results sheet (results_template.csv, column measured):

- SP-1-01 (1): passes the register criterion: no crack, no skin-core disbond, deflection inside the 14 mm standoff. Type Y or N.

Note: T8 (same idea for the rejected d6L) retired

Defined in: designs/pack_5200/print/frame/PRODUCTION.md (QC proof load)

15. T4a: drop-step 2.0 g, 20 drops

Why: HA-LS-1 stumble level

Specimen: P2

load	N	lb	kg on the scale
pack load, 20 drops at 2.0 g	667	150	68.0

Steps

1. 150 lb

2. Lift by haul loop h mm, release on the snap shackle

3. Raise h in 5 mm steps to 2.0 g on phyphox

4. 20 drops

Pass: no damage; if 2.0 g needs < 10 mm, revisit the standard

Write in the results sheet (results_template.csv, column measured):

- T4a-01 (1): passes the register criterion: no damage; if 2.0 g needs < 10 mm, revisit the standard. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

16. T4b: single-foot set-down (+ dial on the branch)

Why: branch buckling

Specimen: P2

Steps

1. 150 lb upright, raised 50 mm, dropped onto one foot on the gym mat, 5 x each foot

2. Dial on the upper branch

Pass: no damage at foot/lumbar, tap clean; record peak g and branch deflection

Write in the results sheet (results_template.csv, column measured):

- T4b-01 (1): passes the register criterion: no damage at foot/lumbar, tap clean; record peak g and branch deflection. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/frame_h1.md §8

17. T4c: empty abuse drop 122 cm at -20 C

Why: cold-weather use

Specimen: P2, empty, bag on

Steps

1. 4 h freezer soak

2. Drop 122 cm onto 19 mm plywood on concrete, 6 orientations

Pass: crown still telescopes and locks, no crack, tap clean

Write in the results sheet (results_template.csv, column measured):

- T4c-01 (1): passes the register criterion: crown still telescopes and locks, no crack, tap clean. Type Y or N.

Note: CONFLICT: load_standard has no temperature; three later docs say -20 C. -20 C used. ASSUMPTION household freezer ~-18 C

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/frame_comp_verdict.md §6; designs/pack_5200/analysis/printed_frame_study.md

18. T5: cycle 0.4 -> 1.6 g, 1e5 cycles (H1 body)

Why: frame release

Specimen: P2

load	N	lb	kg on the scale
dead load	267	60	27.2
peak	1068	240	108.8

Steps

1. 60 lb dead + crank/bungee to 240 lb peak, ~50/min

2. Stop every 25,000: T1(a) re-check + tap

Pass: no crack/debond, stiffness change <= 10 %, crown slip < 1 tooth

Write in the results sheet (results_template.csv, column measured):

- T5-01 (1): passes the register criterion: no crack/debond, stiffness change <= 10 %, crown slip < 1 tooth. Type Y or N.

Note: was 'on the C-9 body'; H1 replaced C-9

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/weight_c_structural.md §7

19. T5f: float US1 share 56 kg

Why: float meat limit 53 -> 56 kg

Specimen: frame + cradle, 56 kg wet sandbags

load	N	lb	kg on the scale
float meat	549	124	56.0

Steps

1. Scale in series with US1

2. Crank 0 -> 1.6 g

3. Record US1 share

Pass: share <= 0.95 x the no-friction value

Write in the results sheet (results_template.csv, column measured):

- T5f-01 (1): passes the register criterion: share <= 0.95 x the no-friction value. Type Y or N.

Note: optional; only if the owner wants 56 kg

Defined in: designs/pack_5200/analysis/float_fea.md §3; designs/pack_5200/analysis/load_standard.md §4.3

20. T2u: ultimate demo 450 lb on P1 (destroys P1)

Why: none after it (last test)

Specimen: P1 frame (sacrificial)

load	N	lb	kg on the scale
ultimate, 30 s	2002	450	204.0

Steps

1. Outdoors, backup sling, exclusion zone

2. Step to 450 lb, hold 30 s

Pass: still carries the load at 30 s; record first noise

Write in the results sheet (results_template.csv, column measured):

- T2u-01 (1): passes the register criterion: still carries the load at 30 s; record first noise. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

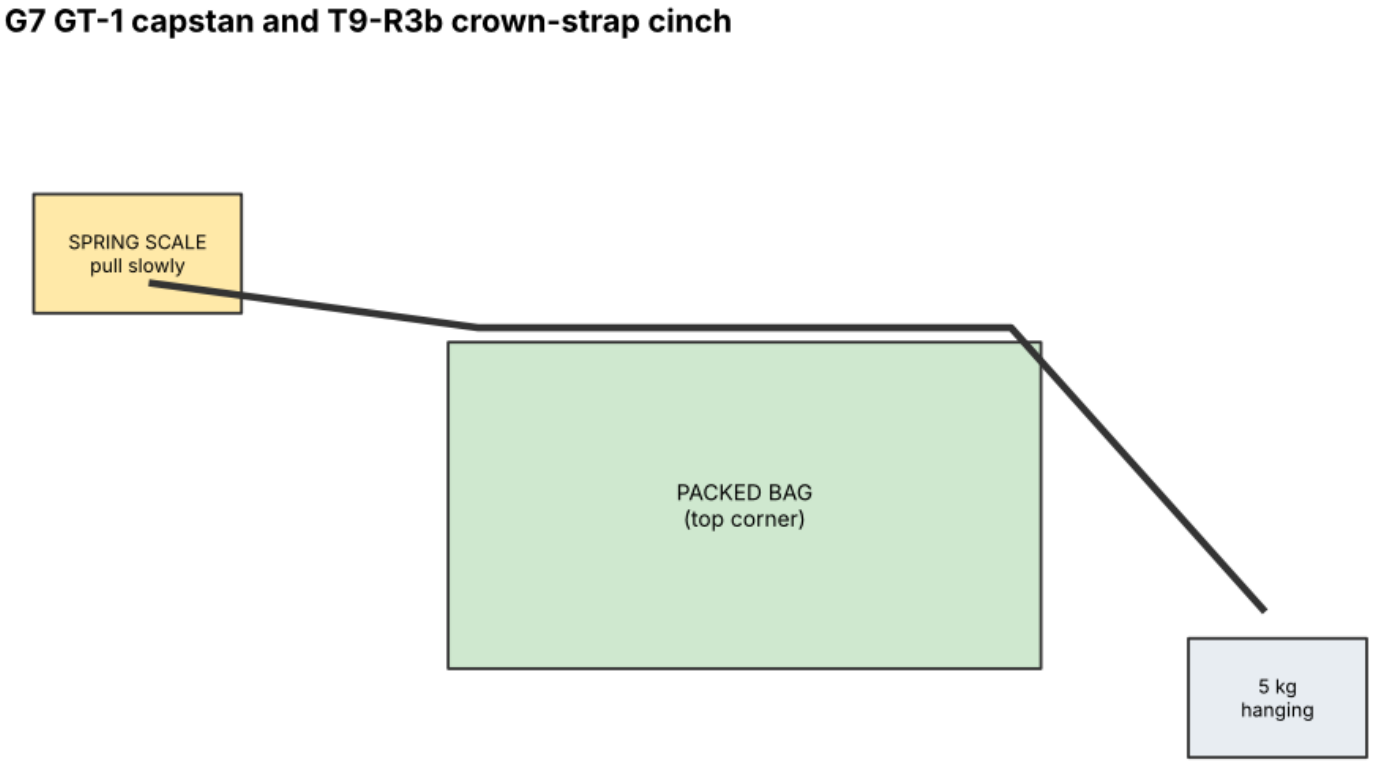
After the session

1. Fill measured and date for every row you ran; leave the rest empty.

2. Photograph every failed or doubtful sample; put the photo number in photo.

3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

G7: Complete pack: fit, field and first-article bag checks



When: after T1-T4 pass

Rig: the complete pack, sand bags, spring scale, FSR strips, phone (phyphox)

Kits: D, E

The bag's batch release (T9-R3b/d, GT-1, PT-7) comes first in this session, then the BL-037 comfort gates, then the field tests.

About 24.9 h hands-on and 2 h of machine time.

Safety (read before you start)

- Loaded walks: a second person nearby, flat ground first, stop at any crack noise. Float meat limit 53 kg (56 kg only after T5f).

Rig sketch

GT-1 capstan: spring scale <-- guy webbing over the packed bag's top corner --> 5 kg hanging

mu = ln(T / 49 N) / wrap angle (rad)

T9-R3b: spring scale on each crown strap tail: pull to 8.2 kg (80 N), lock

Picture: sketches/G7_rig.png

Before: T1, T3, T2 and T4 have passed on P2 (G6). Then the pack may be carried.

Tests in this session, in order

#	test	releases
1	T9-R3b	Bag release, step 2: batch
2	T9-R3d	Bag release, step 2: batch
3	GT-1	Bag release, step 2: batch
4	PT-7	Bag release, step 2: batch
5	WP-4	Bag release, step 2: batch
6	WP-5	Bag release, step 2: batch
7	HS-1	BL-037 weight gates
8	ST-1	BL-037 weight gates
9	HB-1	BL-037 weight gates + Field release and the rest
10	HB-D2	BL-037 weight gates + Field release and the rest
11	HB-3	Field release and the rest
12	HB-D1	Field release and the rest
13	HB-D3	Field release and the rest
14	FT-S2	Field release and the rest
15	MH-1	Field release and the rest
16	MH-2	Field release and the rest
17	MH-3	Field release and the rest
18	R28-1	Field release and the rest

1. T9-R3b: T9 rev 3 (b): crown strap cinch force and retention (CROWN_PRE_N, MU_LID)

Why: CORRIE F rev 3 crown straps (bag batch release)

Specimen: sewn rev 3 bag on P2, lid on, full top; 0-15 kg spring scale

load	N	lb	kg on the scale
cinch	80	18	8.2
after the walk, at least	40	9	4.1

Steps

1. Hook the spring scale on each B-28 tail

2. Pull to 8.2 kg (80 N) and lock

3. Walk 30 min loaded (only after T1-T4 pass)

4. Read the tension again (pull the tail until it just moves)

5. Set the pack down hard from 0.5 m on its base

6. Read again

7. Check the lid is still seated

Pass: after the walk >= 40 N, or the lid still seated on the full top (report both values)

Write in the results sheet (results_template.csv, column measured):

- T9-R3b-01 (L): left strap tension after the walk. Type the value in N.

- T9-R3b-02 (R): right strap tension after the walk. Type the value in N.

- T9-R3b-03 (both): >= 40 N after the walk OR lid still seated. Type Y or N.

Note: BL-038: split out of T9 (item b)

Defined in: designs/pack_5200/analysis/corrie_f_body.md 14.9, 15.6

2. T9-R3d: T9 rev 3 (d): sewn lid mass and seat (LID_SLIDER5_G, LID_THREAD, LID-SEAT)

Why: B-29 lid mass booking (bag batch release)

Specimen: first sewn B-29 lid; 0.01 g pocket scale (lid fits on a 200 g scale)

Steps

1. Weigh the lid alone 3 times

2. Photograph it seated on the packed 6-day bag and the full bag, from the rear 3/4 and the side

Pass: lid 47.8 g +/- 5 g (ledger); lid edge on the hinge-band seam, camo caps over the blades

Write in the results sheet (results_template.csv, column measured):

- T9-R3d-01 (lid): sewn lid mass (ledger booking). G, pass if 42.8 to 52.8.

- T9-R3d-02 (lid): lid seated as drawn (photo). Type Y or N.

Note: BL-038: split out of T9 (item d); mass from the ledger at run time

Defined in: designs/pack_5200/analysis/corrie_f_body.md 14.9, 14.10; huntkit/mass/ledger.py adopted_bag_rows (crown lid)

3. GT-1: capstan pull: guy webbing over the packed bag's corner (MU_WEB_FAB)

Why: guy friction share in the CG runs (bag batch release)

Specimen: first sewn bag, packed full; a 25 mm guy webbing

load	N	lb	kg on the scale
hanging weight	49	11	5.0

Steps

1. Lay the guy webbing over the packed bag's top corner

2. Hang 5 kg (49 N) on one end

3. Measure the wrap angle with phyphox (tilt of the free leg minus tilt of the hanging leg, in degrees)

4. Pull the other end slowly with the spring scale until the weight starts to rise

5. Read the peak T

6. 3 pulls

7. Mu = ln(T / 49) / (angle x pi / 180)

Pass: mu inside 0.20-0.40 on each pull (record the value); outside: tell the pack-designer (CG runs re-checked)

Write in the results sheet (results_template.csv, column measured):

- GT-1-01 to GT-1-03 (#1 to #3): mu = ln(T / 49 N) / wrap angle (rad). -, pass if 0.20 to 0.40.

Note: BL-038: added. Needs the first sewn bag

Defined in: designs/pack_5200/analysis/corrie_f_body.md 13.2, 13.8, 13.9; outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv

4. PT-7: door zip proof

Why: CORRIE door

Specimen: sewn bag

Steps

1. Zip on a 68 kg fill

2. 1.6 g walking load 30 min

3. 50 gloved open/close cycles

Pass: no tooth separation

Write in the results sheet (results_template.csv, column measured):

- PT-7-01 (1): passes the register criterion: no tooth separation. Type Y or N.

Defined in: designs/pack_5200/analysis/corrie_body.md §6

5. WP-4: rain-fly fit

Why: G-2

Specimen: F-01 over the packed shell

Steps

1. Pack the bag, wings closed, pouch on

2. Fit the fly

Pass: covers to the B-05 seam, no gap at the hinge band

Write in the results sheet (results_template.csv, column measured):

- WP-4-01 (1): passes the register criterion: covers to the B-05 seam, no gap at the hinge band. Type Y or N.

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4

6. WP-5: TPU trim tick

Why: B-10 trims Hypalon -> TPU 840D

Specimen: ports, hose frames, cam-cup base

Steps

1. Walk with the trims against webbing

2. Listener at 1 m

Pass: no audible tick at 1 m

Write in the results sheet (results_template.csv, column measured):

- WP-5-01 (1): passes the register criterion: no audible tick at 1 m. Type Y or N.

Note: B-10 is in B.2: a fail adds 12.8 g back

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4

7. HS-1: shoulder pressure: XLPE L200 vs EVA 45C J-strap load foam

Why: BL-037 H-1 (-22.3 g)

Specimen: J-strap pair: left with XLPE L200 5 mm load foam, right with EVA 45C 5 mm (comfort layer, shell, mesh as drawn); 2 FSRs per strap

load	N	lb	kg on the scale
pack, run 1	294	66	30.0
pack, run 2	667	150	68.0

Steps

1. Tape one FSR under each strap at the top of the shoulder and one 60 mm down the front

2. Calibrate each FSR on the puck first

3. 30 kg in the pack, walk 20 min, read the peaks at 5, 10, 15 and 20 min

4. Repeat at 68 kg

5. 3 testers

6. Ask each tester if either strap felt a hard spot (bottoming)

7. Swap sides for the third tester

Pass: XLPE peak / EVA peak <= 1.10 at 30 kg and 68 kg on all 3 testers; no bottoming felt; CS-1 passed

Write in the results sheet (results_template.csv, column measured):

- HS-1-01 to HS-1-06 (T1-30kg to T3-68kg): XLPE peak / EVA peak. -, pass if at most 1.10.

- HS-1-07 to HS-1-09 (T1 to T3): no bottoming felt. Type Y or N.

Note: BL-038: added (BL-037). Walking with 68 kg waits for T1-T4 (frame released)

Defined in: designs/pack_5200/analysis/weight_push.md 2 (H-1), 4, 7 XLPE-SHOULDER

8. ST-1: gloved one-hand sternum clip-on, 25 mm side-release

Why: BL-037 H-2 (-6.0 g)

Specimen: sternum strap with the 25 mm side-release buckle sewn on the harness

load	N	lb	kg on the scale
pack	147	33	15.0

Steps

1. Wear the loaded pack (15 kg)

2. Winter gloves on

3. Clip the sternum buckle with one hand, 10 tries

4. Time each

Pass: 10 of 10 one-handed (record the times)

Write in the results sheet (results_template.csv, column measured):

- ST-1-01 (1): one-hand gloved clip-ons, out of 10. Count, pass if at least 10.

Note: BL-038: added (BL-037); R-06 fit trial session

Defined in: designs/pack_5200/analysis/weight_push.md 2 (H-2)

9. HB-1: hipbelt pressure map, 3 testers, + front-strap sensor

Why: belt B fit (OD-HB1); BL-006 c2 (with HB-D2); front-strap pad (+42.9 g if it fails); BL-037 B-1 / B-3 (XLPE L200 wing load and lumbar foams, 4 mm wing comfort)

Specimen: belt B on 3 testers, 68 kg

Steps

1. FSRs under the belt at the crest and the ASIS, plus one FSR on the front strap 10-20 mm in front of the wing tip under the two 25 mm legs

2. Belt tension T_req read on the scale

3. Record peaks walking and standing

4. If the front-strap sensor fails, repeat with the 110 x 120 mm front-strap pad

Pass: every sensor <= 48.6 kPa (absolute pass, orchestrator call 2026-11-10); bony landmarks watched at 30 kPa; front-strap sensor <= 48.6 kPa without the pad; BL-037 build: wing worst <= 56.5 kPa, lumbar <= 48.6 kPa, hold not worse than belt B on the same testers

Write in the results sheet (results_template.csv, column measured):

- HB-1-01 (T1): wing worst. KPa, pass if at most 56.5.

- HB-1-02 (T1): lumbar worst. KPa, pass if at most 48.6.

- HB-1-03 (T1): front strap. KPa, pass if at most 48.6.

- HB-1-04 (T2): wing worst. KPa, pass if at most 56.5.

- HB-1-05 (T2): lumbar worst. KPa, pass if at most 48.6.

- HB-1-06 (T2): front strap. KPa, pass if at most 48.6.

- HB-1-07 (T3): wing worst. KPa, pass if at most 56.5.

- HB-1-08 (T3): lumbar worst. KPa, pass if at most 48.6.

- HB-1-09 (T3): front strap. KPa, pass if at most 48.6.

Note: crude estimate under the front-strap legs is 79-103 kPa (hipbelt_diet.md), so a fail is likely: then the pad pair (+42.9 g, loadpath_verification §5) is booked and HB-1 re-run with it. Old relative criterion (B <= current - 15 %) dropped; old belt not needed

Defined in: designs/pack_5200/analysis/hipbelt_fit.md §10; designs/pack_5200/analysis/hipbelt_diet.md (front strap); designs/pack_5200/analysis/loadpath_verification.md §5 (front-strap pad), §8 (PAD); docs/DECISIONS.md 2026-11-10 BL-016 and BL-006 rows

10. HB-D2: wing stiffener 0.75 mm roll-over (one wing each)

Why: BL-006 c2 (A_c2); BL-037 B-2 (0.5 mm stiffener)

Specimen: belt B wings side by side: 0.5 mm and 0.75 mm HDPE stiffener (and the 1.0 mm reference)

Steps

1. 68 kg, 30 min hike

2. Mark the crest line on both wings

3. Photograph the top edge at the crest every 10 min

4. HB-1 sensors on both wings

Pass: the thinnest stiffener that passes the HB-D2 roll-over criterion is kept (0.5 mm keeps BL-037 B-2)

Write in the results sheet (results_template.csv, column measured):

- HB-D2-01 (0.5mm): 0.5 mm wing passes roll-over. Type Y or N.

- HB-D2-02 (0.75mm): 0.75 mm wing passes roll-over. Type Y or N.

Note: fail: back to 1.0 mm (+9.8 g)

Defined in: designs/pack_5200/analysis/hipbelt_diet.md (c2, HB-D2)

11. HB-3: hipbelt slip / settle hike

Why: belt B

Specimen: complete pack 68 kg

Steps

1. 30 min hike

2. Mark the crest on the belt

3. Measure creep

Pass: <= 15 mm (ASSUMPTION)

Write in the results sheet (results_template.csv, column measured):

- HB-3-01 (1): passes the register criterion: <= 15 mm (ASSUMPTION). Type Y or N.

Defined in: designs/pack_5200/analysis/hipbelt_fit.md §10

12. HB-D1: no stabilizer straps: side-hill stability

Why: BL-006 b (A_b)

Specimen: complete pack, 68 kg sandbag, stabilizers fitted but slack vs cinched

Steps

1. 20 min side-hill traverse (about 25 deg) and a 10 min step-down set, stabilizers cinched, then fully slack

2. Same tester

3. Belt tension at T_req (luggage scale)

Pass: belt creep within HB-3's 15 mm both ways; no tester-reported roll or sway difference

Write in the results sheet (results_template.csv, column measured):

- HB-D1-01 (1): passes the register criterion: belt creep within HB-3's 15 mm both ways; no tester-reported roll or sway difference. Type Y or N.

Note: fail: refit the straps (+22.4 g); the frame anchor pads stay

Defined in: designs/pack_5200/analysis/hipbelt_diet.md (b, HB-D1)

13. HB-D3: pad-seat cord tie and 1000D sleeve retention

Why: BL-006 f1 + f2 (A_f)

Specimen: belt on the frame, pad seat with cord tie + cord lock and 1000D sleeve

Steps

1. Swing the pack off and set it down 20 times

2. Then hang the pack by the belt from the lumbar pad for 1 min with 10 kg in the bag

Pass: pad stays seated, cord lock holds, sleeve does not fray at the flange edge

Write in the results sheet (results_template.csv, column measured):

- HB-D3-01 (1): passes the register criterion: pad stays seated, cord lock holds, sleeve does not fray at the flange edge. Type Y or N.

Note: fail: strap + ladder lock and Hypalon sleeve back (+12.6 g)

Defined in: designs/pack_5200/analysis/hipbelt_diet.md (f1, f2, HB-D3)

14. FT-S2: no wings: 5 km side-hill + door cycles with the tripod loaded

Why: side wings deleted (A_wings)

Specimen: complete pack, no wings, tripod + spotter pod under W1/W2

Steps

1. Paint-pen mark tripod and pod against the gusset

2. 5 km of side-hill, both slope directions, measure shift every 1 km

3. Then open and close the door 5 times with the tripod loaded (unhook W1/W2, open, close, re-hook) and measure

Pass: shift <= 20 mm throughout; the tripod is never freed by a door cycle

Write in the results sheet (results_template.csv, column measured):

- FT-S2-01 (1): passes the register criterion: shift <= 20 mm throughout; the tripod is never freed by a door cycle. Type Y or N.

Note: door cycles added from BL-014 S7. Fail: a gusset tripod keeper (team B A2 32.2 g; BL-022 proposes a shock-cord keeper) or the wings back (+101.6 g)

Defined in: designs/pack_5200/analysis/strap_architecture.md (ASSUMPTION W1/W2); designs/pack_5200/analysis/corrie_f_body.md; docs/DECISIONS.md 2026-11-11 owner row; designs/pack_5200/analysis/review_20261111.md S7

15. MH-1: bag-on meat mode walk

Why: R-09

Specimen: complete pack

Steps

1. 30 kg sandbags in game bags high on B-01, IM1-IM3 cinched

2. 30 min on a slope

3. Time it gloved

Pass: no load shift you can feel; <= 60 s

Write in the results sheet (results_template.csv, column measured):

- MH-1-01 (1): passes the register criterion: no load shift you can feel; <= 60 s. Type Y or N.

Defined in: designs/pack_5200/BUILD_PLAN.md §5a.2

16. MH-2: frame-only meat haul (float / bag-off)

Why: R-08, float meat limit 53 kg

Specimen: frame + cradle, 36 kg sandbag quarter

Steps

1. IM1-IM3 + wraps

2. Walk 30 min

3. Measure slip

Pass: < 25 mm slip after 30 min

Write in the results sheet (results_template.csv, column measured):

- MH-2-01 (1): passes the register criterion: < 25 mm slip after 30 min. Type Y or N.

Note: stubs in step 1 became fairlead eyes; use the eyes

Defined in: designs/pack_5200/BUILD_PLAN.md §5a.4

17. MH-3: meat strap wash

Why: R-12

Specimen: IM straps

Steps

1. Unhook, machine-wash cold, dry

Pass: hooks and cams still work

Write in the results sheet (results_template.csv, column measured):

- MH-3-01 (1): passes the register criterion: hooks and cams still work. Type Y or N.

Defined in: designs/pack_5200/BUILD_PLAN.md §5a.5; designs/pack_5200/DESIGN.md R-12

18. R28-1: head-clearance photo

Why: R-28

Specimen: worn pack, collar unrolled

Steps

1. Side photo looking up

Pass: ~30 deg tilt before contact

Write in the results sheet (results_template.csv, column measured):

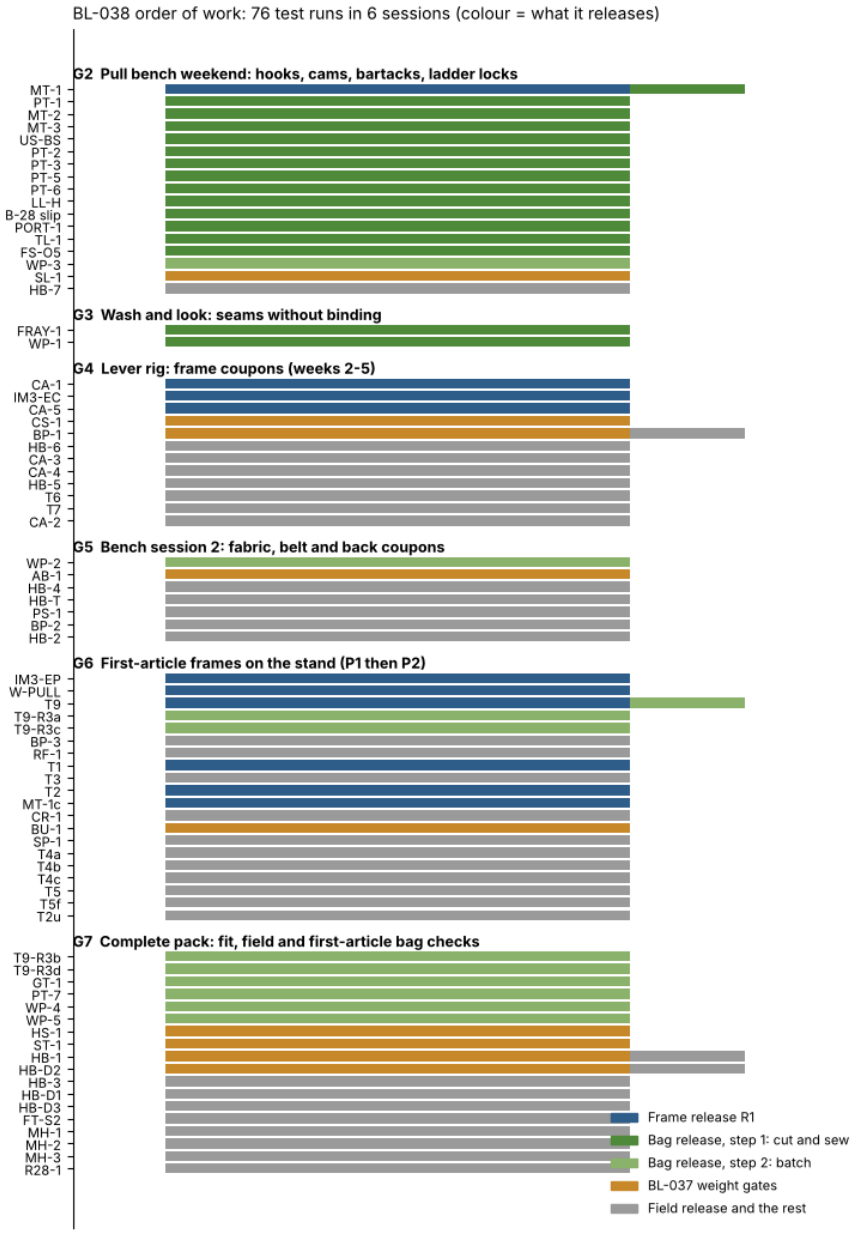
- R28-1-01 (1): passes the register criterion: ~30 deg tilt before contact. Type Y or N.

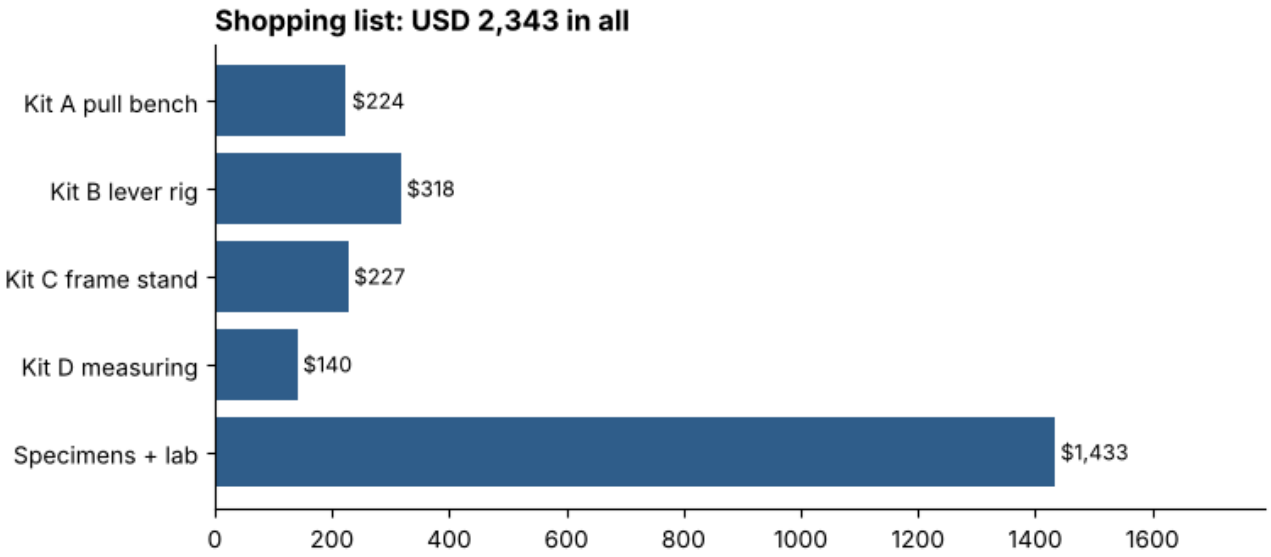
Defined in: designs/pack_5200/BUILD_PLAN.md §5a.6; designs/pack_5200/DESIGN.md R-28

After the session

1. Fill measured and date for every row you ran; leave the rest empty.
2. Photograph every failed or doubtful sample; put the photo number in photo.
3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

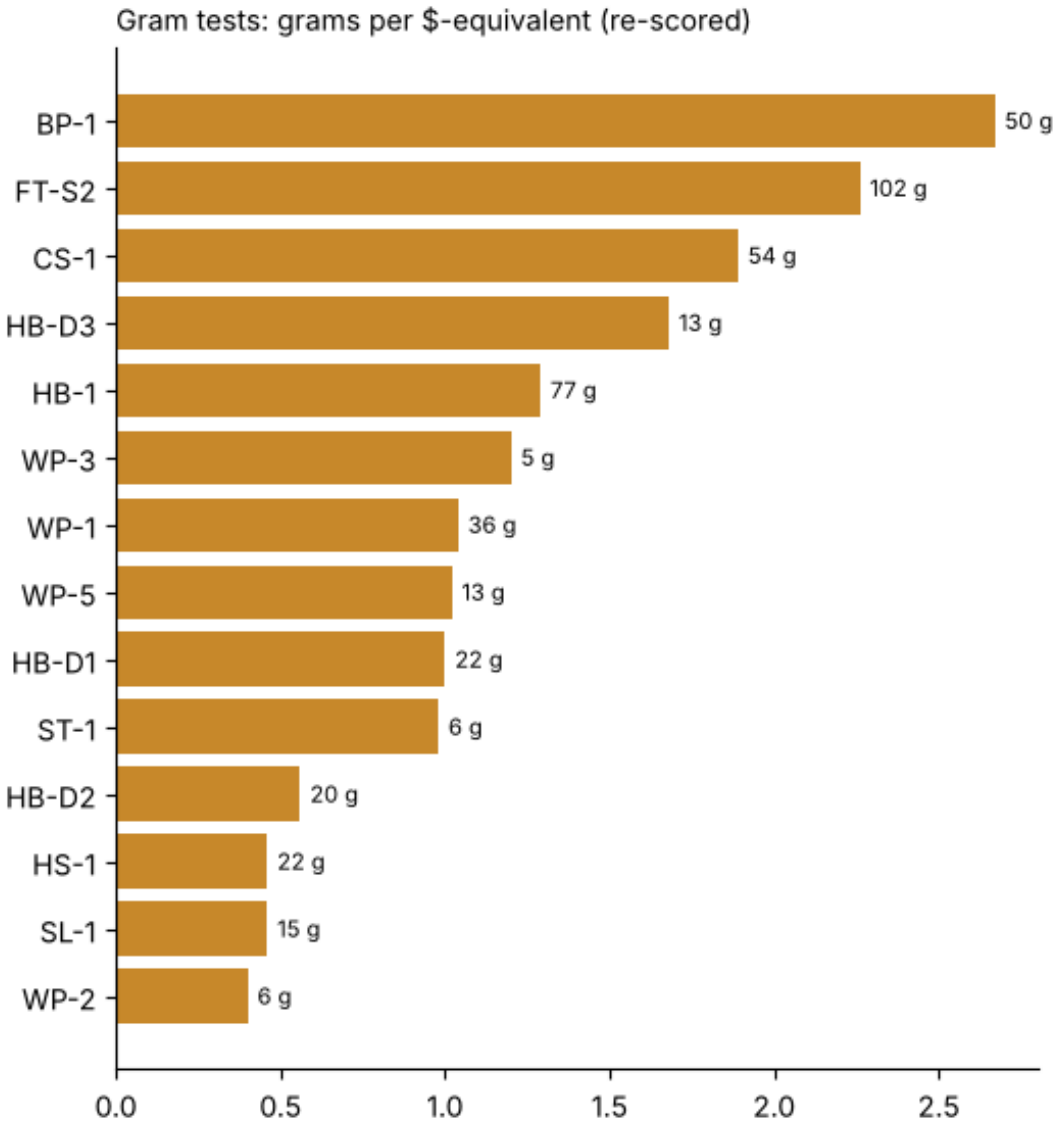
Pictures


BL-038 order of work: 76 test runs in 6 sessions (colour = what it releases)
G2 Pull bench weekend: hooks, cams, bartacks, ladder locks
MT-1
PT-1
MT-2
MT-3
US-BS
PT-2
PT-3
PT-5
PT-6
LL-H
B-28 slip
PORT-1
TL-1
FS-O5
WP-3
SL-1
HB-7
G3 Wash and look: seams without binding
FRAY-1
WP-1
G4 Lever rig: frame coupons (weeks 2-5)
CA-1
IM3-EC
CA-5
CS-1
BP-1
HB-6
CA-3
CA-4
HB-5
T6
T7
CA-2
G5 Bench session 2: fabric, belt and back coupons
WP-2
AB-1
HB-4
HB-T
PS-1
BP-2
HB-2
G6 First-article frames on the stand (P1 then P2)
IM3-EP
W-PULL
T9
T9-R3a
T9-R3c
BP-3
RF-1
T1
T3
T2
MT-1c
CR-1
BU-1
SP-1
T4a
T4b
T4c
T5
T5f
T2u
G7 Complete pack: fit, field and first-article bag checks
T9-R3b
T9-R3d
GT-1
PT-7
WP-4
WP-5
HS-1
ST-1
HB-1
HB-D2
HB-3
HB-D1
HB-D3
FT-S2
MH-1
MH-2
MH-3
R28-1
Frame release R1
Bag release, step 1: cut and sew
Bag release, step 2: batch
BL-037 weight gates
Field release and the rest



ORDER FIRST (long lead):

- cam buckle 25 mm aluminium (Austere type) (MT-3, US-BS)
  LONG LEAD: RBTR all variants sold out, Extremtextil stock problems (2026-11-13). Order now; buy the pack's 5 cams in the same order
- YKK #10 coil slider, metal (cord pull fitted) (SL-1)
  LONG LEAD: Rockywoods listed sold out (2026-11-13); Shelby Outdoor EU stocks the YKK 10C slider
- ROHACELL 71 IG-F core for the 2 mm strip (BP-1)
  LONG LEAD: 2 mm is not a stock sheet (3 mm thinnest, back_system_integration.md); sand a 3 mm offcut or special order
- HyperD 300 printable, custom camo print (AB-1, FRAY-1)
  LONG LEAD: custom print, lead time not published
- Martindale abrasion test, accredited lab (2 samples + order fee) (AB-1)
  LONG LEAD: about 4 weeks after the samples arrive
- sacrificial P1 frame (IM3-EP, W-PULL, then T2u) (IM3-EP, W-PULL, T2u)
  print time: allow 2 weeks with P2



Frame release R1

CA-1 release level: 0.276
governing flip 0.2556
(body 0.2553, 28 mm cheek
0.2464 / 0.2556 at +10 deg)
+ 0.02

MT-1c swing sets the level:
<= 0.4 %: 0.276
<= 70 %: 0.40
above: 0.404

CA-1 at 0.40 is worth 9.9 g
(option F-1, not booked; was 60 g)
rank 18 of 20